Highlights

Multi-Agent Reinforcement Learning for Surface–Subsurface Observation of Tropical Cyclones Using Submersible Unmanned Vessels

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- Dual-mode SUVs track surface winds and sample subsurface data.
- Historical atmosphere–ocean environments cover 500 TCs in the western North Pacific.
- Parameter-sharing IPPO is trained for TC observation with 4–16 SUVs.
- Compared with traditional path-planning algorithms at both stages.

Multi-Agent Reinforcement Learning for Surface–Subsurface Observation of Tropical Cyclones Using Submersible Unmanned Vessels

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ABSTRACT

Tropical cyclones are destructive, while intensity and rapid-intensification forecasts remain limited by sparse near-core observations. Conventional platforms rarely combine rapid transit with surface-wind and subsurface-current sensing. Blue Whale-class submersible unmanned vessels can transit rapidly at the surface to TC cores, then dive to observe temperature, salinity, currents, and biogeochemical variables. We use simulation-based multi-agent reinforcement learning for autonomous observation and navigation. The environments combine ERA5 10 m winds and HYCOM currents from 0 to 60 m with IBTrACS tracks for 500 western North Pacific TCs. A parameter-sharing independent proximal policy optimization (IPPO) policy maps shared surface-wind observations or local current histories to continuous control commands at 10 min intervals. Surface policies use shared wind observations; subsurface policies use only local current histories. We compare 12 m/s surface and 1.5 m/s submerged limits, close to current Blue Whale design speeds, with 20 m/s and 2.0 m/s as future upgrades. Comparison between the four traditional planners and IPPO at surface experiments shows that the IPPO policy achieves the smallest final fleet-to-center distance, which means that the SUVs can reach the TC core region more readily and obtain more valuable observations. For the surface IPPO policy, we further evaluate ring, line, grid, and V-shaped formations. We assume that after reaching the TC core region, the SUVs dive vertically to different depths from 0 to 60 m for further observations. Underwater, IPPO attains the largest geometric and current-weighted coverage, while the potential field algorithm attains the highest mean experienced current. Limitations remain that training and validation are conducted in a simulation environment built from historical reanalysis data; real-vehicle experiments are not included and require future work.

1. Introduction

Tropical cyclones (TCs) threaten coastal communities, infrastructure, and offshore operations. Climate assessments indicate that some TC changes are detectable despite observational uncertainty [1] and project an increasing proportion of very intense TCs under continued warming [2], while global TC frequency may decline [3] and rapid-intensification events have recently increased in offshore regions [4]. Landfalling TCs over the western North Pacific and China have also become more destructive, with increasing nearshore intensification and exposure [5, 6, 7, 8]. Track forecasts have improved substantially, but intensity prediction remains more difficult: recent intensity guidance has reduced errors [9], yet official forecasts in the western North Pacific continue to carry substantial uncertainty [10]. A central reason is that intensity depends on air–sea enthalpy and momentum exchange, upper-ocean heat content, barrier layers, mesoscale eddies, and TC-induced currents [11, 12].

Existing observing platforms each cover only part of this problem. Spaceborne microwave sensors provide broad but complementary and incomplete views of the near-core wind field. Active scatterometers infer surface wind from ocean backscatter, but lose sensitivity at high wind speeds and can be degraded by rain-induced attenuation and scattering [13]. Passive radiometers infer wind from ocean emissivity and can penetrate some cloud cover, but cloud liquid water, precipitation emission, and atmospheric attenuation contaminate the brightness-temperature signal [13, 14]. Neither sensor class resolves the subsurface ocean [15]. Aircraft obtain high-quality near-core measurements but are expensive and campaign-limited [16, 17], whereas gliders remain at sea for long periods but move much more slowly than a translating TC [18]. AUVs can adapt their sampling to changing environmental and vehicle states [19]. Upper-ocean observations can improve initial states and, in some cases, intensity forecasts [20, 21, 22, 23, 24], but their value is highly sensitive to sampling location and background structure. Targeted observation studies likewise show that forecast benefit depends on sensitive regions, the background state, and the sampling design [25, 26, 27]. These limitations leave near-core wind and upper-ocean states poorly constrained in many forecast initial conditions. Adaptive TC observation requires

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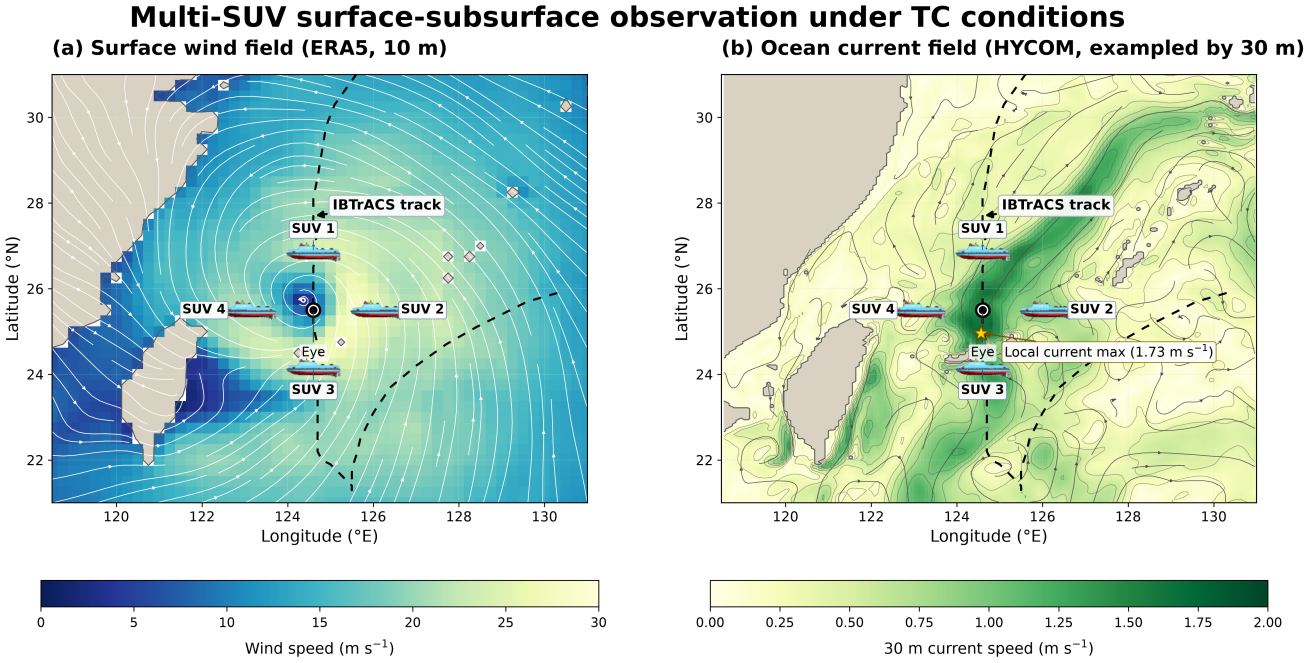


Figure 1: Coordinated surface–subsurface TC observation for a representative HINNAMNOR (2022) snapshot. The left panel shows four SUVs tracking the IBTrACS best track while sampling ERA5 winds at 10 m above the sea surface. The right panel shows the same formation sampling HYCOM currents at 30 m below the sea surface; this depth is shown as an illustrative example, whereas the complete experiments cover the seven depths from 0 to 60 m. The four-vehicle configuration is illustrative, with equal angular spacing; subsequent experiments evaluate fleets of 8, 12, and 16 vehicles.

decisions about motion, sampling time, and coordination, not coverage alone.

A submersible unmanned vessel (SUV) combines rapid surface transit with subsurface sensing [28, 29, 30, 31]. The operational concept assumes that the SUVs are deployed from a mother ship, such as a large research vessel or an equivalent offshore support ship, or from an offshore floating platform such as the Shanghai Jiao Tong University Far-Seas Floating Island (Yuanhai Fudongdao). Such platforms can provide rapid relocation, long-duration station keeping, and deployment of heavy marine equipment [32, 33, 34]. The Blue Whale is used as an engineering reference: its public specifications indicate a surface speed of about 23 kn (11.83 m/s), a submerged speed of about 3 kn (1.54 m/s), and an observational payload that includes an acoustic Doppler current profiler and a CTD sensor, with a public reference depth of approximately 60 m [28]. This study uses the current Blue Whale's surface–subsurface motion capabilities as the reference parameterization and evaluates a possible future-upgrade configuration as a comparison case. The vehicles are simplified as point masses for path planning. In the proposed mission, the surface fleet observes 10 m winds and follows the moving TC center. Final tracking error measures the remaining fleet-to-center separation. At the subsurface, four SUVs are assigned to each prescribed layer and seek the strongest observable current. Figure 1 illustrates this surface-to-subsurface task switch; four vehicles are shown at the surface as an example, and the same parameter-sharing policy is later evaluated with 4, 8, 12, and 16 SUVs.

Traditional graph search, sampling-based planning, potential fields, and model predictive control are effective when the cost map and dynamics are known [35, 36, 37, 38, 39, 40]. In a TC, however, the fields are nonstationary and only locally observed, while propulsion is superimposed on ambient flow. Conventional pipelines also estimate a field or waypoint sequence before execution. By contrast, reinforcement learning (RL) maps each new measurement directly to the next action, closing the observation–decision–motion loop. RL has demonstrated strong sequential decision making across diverse domains [41, 42, 43, 44, 45, 46, 47, 48], and independent PPO is competitive in cooperative multi-agent tasks [49, 50, 51]. Multi-AUV adaptive sampling and target tracking provide the ocean-robotics foundations [52, 53, 54, 55, 56].

Three gaps remain. First, a unified atmosphere–ocean reinforcement-learning environment for coordinated surface–subsurface TC observation has not been established [57, 58]. Second, it remains unclear whether parameter-sharing MARL can track a moving TC under deployment uncertainty, sample the subsurface current field, and remain effective across different fleet sizes [25, 26, 27, 58]. Third, learned policies have rarely been compared with causal planners under identical local observations, motion constraints, and control interfaces. We formulate stochastic policy optimization under partial observability [59, 49, 50], evaluate 4-, 8-, 12-, and 16-SUV fleets, and provide this matched comparison.

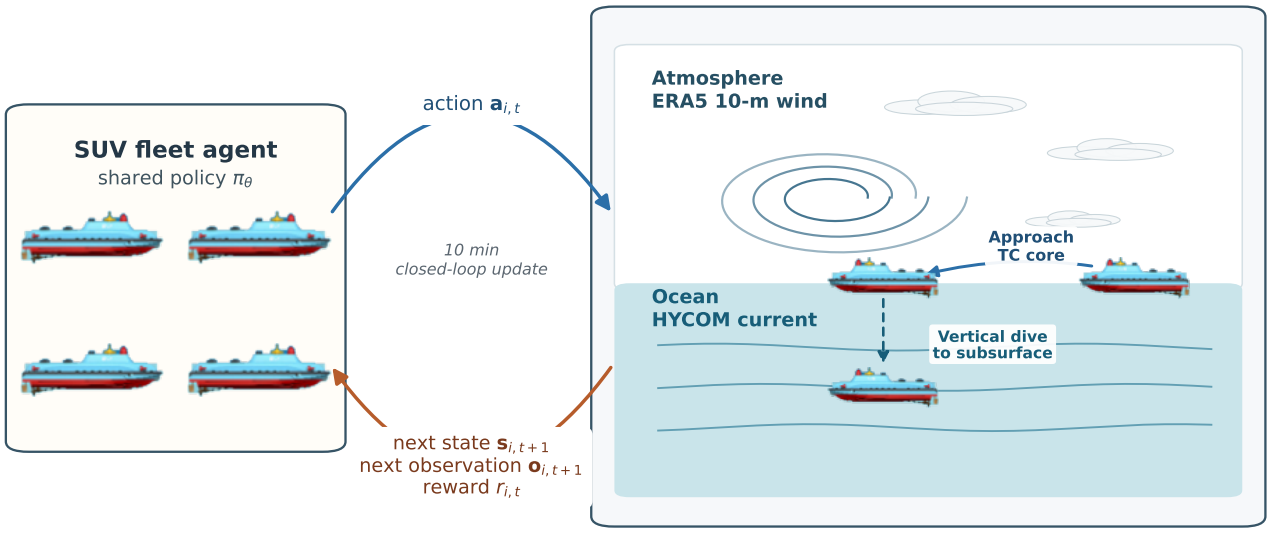


Figure 2: Closed-loop interaction between the SUV fleet agent and the tropical-cyclone atmosphere–ocean environment. At each 10 min interval, the SUV fleet agent perceives environmental information as observations. Its shared policy π_θ maps these observations to actions and sends them to the environment, which returns a reward and the next state to the agent. In the right panel, the rightmost surface SUV senses the local wind speed and moves toward the TC core region. After reaching a more central position, represented by the left surface SUV, the first-stage task is completed. The SUV then dives vertically and begins the second-stage observation task at the prescribed depth, collecting temperature, salinity, and current observations and intelligently adjusting its motion based on the local current information.

The main contributions of this study are summarized as follows.

1. We construct a unified atmosphere–ocean RL environment by aligning ERA5 10 m winds, HYCOM currents at seven depths from 0 to 60 m, and IBTrACS tracks. The 500-TC archive is divided at the TC level into 450 training TCs and 50 held-out evaluation TCs.
2. We evaluate a parameter-sharing IPPO policy with globally shared surface-wind observations and decentralized subsurface current histories for 4-, 8-, 12-, and 16-SUV fleets. Surface policies are tested at 100–800 km deployment offsets, and subsurface policies are tested at seven depths and two propulsion limits. We report final and mean tracking error, experienced current, geometric coverage, and current-weighted coverage.
3. We compare IPPO with four traditional planners, including artificial potential field, receding-horizon A*, lawnmower coverage, and information-greedy control, under matched local observations, action constraints, motion constraints, control intervals, and paired evaluation episodes.

2. Problem Formulation

2.1. Observation Task and Design Requirements

The objective is to coordinate a fleet of submersible unmanned vessels (SUVs) to observe a moving tropical cyclone (TC) across the sea surface and upper ocean. At the surface, each SUV navigates toward the moving TC center. Final tracking error measures the remaining fleet-to-center separation and serves as a proxy for access to the near-core high-wind regime. In the operational concept, the fleet would descend after entering a predefined terminal tracking region. The learning experiments evaluate this mission as two separate episodes: a surface approach episode and a subsurface sampling episode in which four SUVs seek the highest attainable current speed at each prescribed depth. The mission imposes four operational requirements:

1. rapid correction of initial deployment errors using surface propulsion;
2. decentralized tracking of the moving TC using onboard wind observations and inter-vehicle observation sharing;
3. subsurface observation at ocean depths of 0, 10, 20, 30, 40, 50, and 60 m below the sea surface;
4. safe operation under action smoothing and coastline safeguards.

The surface maximum speed is varied between 12 and 20 m/s, while the submerged maximum propulsion speed is 1.5 or 2.0 m/s. This difference creates an asymmetric decision problem: surface propulsion is used to follow the TC and reduce the fleet-to-center distance, quantified primarily by the final tracking error. At the subsurface, the primary learning objective changes to finding and sampling higher-current water, while geometric and current-weighted spatial coverage characterize how broadly and how valuably the

fleet samples the dynamically active region. The four SUVs must balance active propulsion against passive advection by the ambient current.

2.2. Problem Definition

The fleet contains N SUVs, where $N \in \{4, 8, 12, 16\}$, indexed by $i \in \{1, 2, \ldots, N\}$. The position, commanded horizontal velocity, and operating mode of SUV i at time t are denoted by $\mathbf{x}_i(t)$, $\mathbf{p}_i(t)$, and $m_i(t) \in \{\mathrm{s}, \mathrm{u}\}$, respectively, where s and u denote the surface and subsurface modes. Horizontal positions and reported distances are expressed in km, whereas commanded and ocean-current velocities are expressed in m/s. The commanded speed satisfies

$$\|\mathbf{p}_i(t)\| \le \begin{cases} V_{\max}^{\mathrm{s}}, & m_i(t) = \mathrm{s}, \\ V_{\max}^{\mathrm{u}}, & m_i(t) = \mathrm{u}. \end{cases} \tag{1}$$

Each simulated episode is assigned to one stage, consistent with the separate surface and subsurface environments used in training and evaluation. The mode $m_i(t)$ selects the applicable speed bound and motion law for that episode; continuous diving, ballast transitions, and resurfacing are not simulated within a single decision trajectory.

For the motion equations, let $\mathbf{p}_i$ denote the desired horizontal velocity command and $\mathbf{q}_i^{\mathrm{prop}}$ the effective propulsion velocity after the acceleration and heading dynamics. The km/min representation of the ocean current is denoted by $\mathbf{v}_{\mathrm{ocean}}$; the device-level speed bounds above remain reported in m/s. The surface uses the effective propulsion velocity directly as ground velocity, whereas the underwater stage adds current advection. The horizontal motion in km is

$$\dot{\mathbf{x}}_i(t) = \begin{cases} \mathbf{q}_i^{\mathrm{prop}}(t), & m_i(t) = \mathrm{s}, \\ \mathbf{q}_i^{\mathrm{prop}}(t) + \mathbf{v}_{\mathrm{ocean}}\left(\mathbf{x}_i(t), z_i, t\right), & m_i(t) = \mathrm{u}, \end{cases} \tag{2}$$

where $z_i \in \{0, 10, 20, 30, 40, 50, 60\}$ m. Time t is measured in minutes, and $\mathbf{v}_{\mathrm{ocean}}(\cdot)$ is the HYCOM current at the corresponding position and depth. Underwater ground velocity is the superposition of active propulsion and background advection.

For numerical implementation, the continuous model is integrated explicitly at the decision interval $\Delta t = 10$ min:

$$\mathbf{x}_{i,k+1} = \mathbf{x}_{i,k} + \Delta t \begin{cases} \mathbf{q}_{i,k}^{\mathrm{prop}}, & m_{i,k} = \mathrm{s}, \\ \mathbf{q}_{i,k}^{\mathrm{prop}} + \mathbf{v}_{\mathrm{ocean}}(\mathbf{x}_{i,k}, z_i, t_k), & m_{i,k} = \mathrm{u}. \end{cases} \tag{3}$$

Here $\Delta t = 10$ min. The current field is evaluated at the beginning of each interval and interpolated to the current vehicle position. For the active underwater reward, we write this pre-move current as $\mathbf{u}_{i,k} = \mathbf{u}_{\mathrm{ocean}}(\mathbf{x}_{i,k}, z_i, t_k)$.

To represent the dynamic constraints of the SUV, the desired command is converted into a speed and course and then tracked under bounded acceleration, deceleration, heading rate, and minimum turn radius. In the surface stage, the command specifies a desired horizontal ground velocity, which is tracked without an added current term. In the underwater stage, the command specifies a desired propulsion velocity before current advection is applied:

$$\begin{aligned} \mathbf{p}_{i,k}^{\mathrm{des}} &= V_{\max}^{m_i} \bar{\mathbf{a}}_{i,k}, \qquad s_{i,k}^{\mathrm{des}} = \left\| \mathbf{p}_{i,k}^{\mathrm{des}} \right\|_2, \\ \psi_{i,k}^{\mathrm{des}} &= \operatorname{atan2}\left(\bar{a}_{i,k,y}, \bar{a}_{i,k,x}\right). \end{aligned} \tag{4}$$

Each SUV carries an effective propulsion speed s_i and heading ψ_i, with $\mathbf{q}_i^{\mathrm{prop}} = s_i[\cos\psi_i, \sin\psi_i]^\top$. The 10 min control interval is divided into ten 60 s substeps. In substep ℓ, speed and heading evolve according to

$$\begin{aligned} s_i^{\ell+1} &= \mathrm{clip}\big(s_i^{\ell} \\ &\quad + \mathrm{clip}\Big(s_i^{\mathrm{des}} - s_i^{\ell}, -a_{m_i}^{\mathrm{dec}}\Delta\tau, a_{m_i}^{\mathrm{acc}}\Delta\tau\Big), 0, V_{\max}^{m_i}\Big), \\ \bar{\omega}_i^{\ell+1} &= \min\left(\omega_{\max}^{m_i}, \frac{s_i^{\ell+1}}{R_{\min}^{m_i}}\right), \\ \psi_i^{\ell+1} &= \mathrm{wrap}\big[\psi_i^{\ell} \\ &\quad + \mathrm{clip}\big(\mathrm{wrap}\left(\psi_i^{\mathrm{des}} - \psi_i^{\ell}\right), -\bar{\omega}_i^{\ell+1}\Delta\tau, \bar{\omega}_i^{\ell+1}\Delta\tau\big)\big], \end{aligned} \tag{5}$$

where $\Delta\tau = 60$ s. For the surface stage, $a_{\mathrm{s}}^{\mathrm{acc}} = 0.02$ m/s^2, $a_{\mathrm{s}}^{\mathrm{dec}} = 0.03$ m/s^2, $\omega_{\max}^{\mathrm{s}} = 1°$/s, and $R_{\min}^{\mathrm{s}} = 200$ m. For the underwater stage, $a_{\mathrm{u}}^{\mathrm{acc}} = 0.002$ m/s^2, $a_{\mathrm{u}}^{\mathrm{dec}} = 0.003$ m/s^2, $\omega_{\max}^{\mathrm{u}} = 2°$/s, and $R_{\min}^{\mathrm{u}} = 100$ m. The minimum-radius term is applied only while the vehicle is moving; a commanded zero action decelerates the vehicle without changing its course. The vehicle then advances over each substep using

$$\mathbf{x} \leftarrow \mathbf{x} + \left(\mathbf{q}_i^{\mathrm{prop}} + \mathbf{1}_{\{m_i = \mathrm{u}\}}\mathbf{u}_{\mathrm{ocean}}\right)\Delta\tau, \tag{6}$$

with the current held fixed over the 10 min control interval. Thus the surface desired command is a ground velocity, whereas the underwater desired command is a propulsion velocity before current advection. The speed is initialized to zero and the heading is initialized uniformly in $[0, 2\pi)$ at the beginning of each episode.

The motion model remains a horizontal point mass model and does not resolve added mass, hydrodynamic drag, roll and pitch, buoyancy and ballast dynamics, thruster transients, wave loads, or actuator delay. ERA5 winds are available at 0.25° and hourly resolution, whereas HYCOM currents are available at approximately 0.04° by 0.08° and three-hourly resolution. These fields are interpolated to the 10 min decision interval and to each vehicle position, but such interpolation does not recover high-frequency wave forcing, local wake effects, or unobserved vehicle attitude and actuator states. A detailed six-degree-of-freedom model would require platform-specific hydrodynamic coefficients and forcing information that are not commensurate with the present data, and could imply a level of numerical precision unsupported by the inputs. The results therefore address planning and coordination rather than structural loading or low-level controller response.

2.3. Surface–Subsurface Two-Stage Task

The SUVs first perform surface tracking, using coordinated propulsion to follow the TC and reduce their distance to its center. For the operational protocol, we assume that descent begins after the surface fleet enters the terminal region defined by a final tracking error of at most 200 km. The simulation uses this 200 km value as the upper bound on the tracking error at the surface-to-subsurface transition and initializes the subsurface vehicles within the corresponding 200 km disk around the TC center at the end time of the surface segment. Because the present simulation does not model descent dynamics, the surface and subsurface stages are represented as two time-aligned but state-independent episodes. The descent duration is simplified: after the surface stage, each vehicle is assumed to arrive immediately at one prescribed depth. Each depth is a separate configuration with four SUVs, and a single episode does not visit multiple depths. At the end of the underwater observation task, the operational concept returns the SUVs to the sea surface for recovery. This ascent and recovery phase is not modeled here; the paper models only the surface observation-and-tracking stage and the subsequent fixed-depth underwater observation stage.

The local observation of SUV i is

$$\mathbf{o}_i(t) = \begin{cases} \left[\mathbf{s}_i(t),\, \mathbf{v}_i(t)\right], & m_i(t) = \mathrm{s}, \\ \left[\mathbf{h}_i(t),\, \mathbf{v}_i(t)\right], & m_i(t) = \mathrm{u}, \end{cases} \tag{7}$$

where $\mathbf{s}_i(t)$ is the stage-specific measurement block, $\mathbf{v}_i(t)$ is the normalized effective horizontal propulsion velocity $\mathbf{v}_i = \mathbf{q}_i^{\mathrm{prop}}/V_{\max}^{m_i(t)}$, and $\mathbf{h}_i(t)$ is an optional causal history block. In the final surface experiments, $\mathbf{s}_i(t) \in \mathbb{R}^{3N}$ contains the normalized speed and two wind components at 10 m above the sea surface sampled by all N SUVs; the surface policy has no history block, and its effective ground velocity is augmented by $[\cos\psi_i, \sin\psi_i]$. Surface operation assumes idealized, lossless, and instantaneous inter-vehicle communication: every SUV receives the wind observations acquired by all other SUVs in the fleet as part of its policy input. In the underwater IPPO evaluation, no inter-vehicle communication is assumed. The observation contains no separate instantaneous-current block in addition to the history; instead, define the local current feature $\mathbf{c}_{i,t} = [\|\mathbf{u}_{\mathrm{ocean}}(\mathbf{x}_{i,t}, z_i, t)\|, u_{\mathrm{ocean},x}, u_{\mathrm{ocean},y}]^\top$, where $\mathbf{u}_{\mathrm{ocean}} = (u_{\mathrm{ocean},x}, u_{\mathrm{ocean},y})$, and $\mathbf{h}_i(t) = [\mathbf{c}_{i,t-4}^\top, \ldots, \mathbf{c}_{i,t}^\top]^\top \in \mathbb{R}^{15}$, which is the causal five-frame history of SUV i's current speed and two components at its assigned depth; before five measurements are available, the earliest observed frame is repeated to fill the history. Each underwater SUV receives neither the current observations nor the positions or actions of the other SUVs. The complete reanalysis field, the global best-track center, and future environmental states are not provided to the policy. These two communication assumptions are intentionally different and mark the bounds of the present study. The idealized global surface link represents a favorable satellite or radio-connected operating mode, which makes it an upper-bound assumption. The no-communication underwater case is the conservative opposite bound: shared policy parameters are loaded separately on each vehicle, but no observation, position, or action is exchanged at runtime. The surface payload is $3N$ scalar wind values per vehicle per decision, whereas the underwater inter-vehicle message payload is zero. The reported experiments do not model bandwidth, packet loss, acoustic latency, or intermittent surfacing. We do not present either the ideal surface link or the fully decentralized subsurface case as a validated communication system. The resulting surface observation dimension is $d_o^{\mathrm{s}} = 3N+4$. The underwater five-frame history contains 15 scalars; adding the two normalized propulsion components and the heading representation $[\cos\psi_i, \sin\psi_i]$ gives $d_o^{\mathrm{u}} = 19$.

During training, the reward function can access the distance between each SUV and the best-track TC center. This distance supervises the association between local 10 m wind patterns and proximity to the TC center/high-wind region; the center coordinate itself is not an actor input. During validation, rewards do not influence actions. The trained policy receives only the shared fleet-level wind samples and must track an unseen TC, thereby testing whether the learned wind-to-center relation generalizes.

The normalized action is

$$\mathbf{a}_i(t) \in [-1, 1]^2, \tag{8}$$

and is projected onto the unit disk before being mapped to a desired velocity command in both modes:

$$\tilde{\mathbf{a}}_i(t) = \frac{\mathbf{a}_i(t)}{\max\left(1, \|\mathbf{a}_i(t)\|_2\right)}, \qquad \mathbf{p}_i(t) = V_{\max}^{m_i(t)} \bar{\mathbf{a}}_i(t). \tag{9}$$

This mapping ensures that the desired and effective speeds never exceed the mode-specific speed limit. Here $\bar{\mathbf{a}}_i(t)$ is the executed normalized command after action smoothing, unit-disk projection, and any stage-specific safety projection; without a safety intervention, $\bar{\mathbf{a}}_i(t) = \tilde{\mathbf{a}}_i(t)$. In the surface stage, the action specifies a desired ground velocity. In the underwater stage, the same unit-disk command specifies a desired propulsion speed and course. In both stages, the effective speed and heading evolve according to the dynamics defined above.

2.4. Cooperative Dec-POMDP Formulation

The cooperative task is formulated as a decentralized partially observable Markov decision process (Dec-POMDP):

$$\mathcal{M} = \langle \mathcal{I}, \mathcal{S}, \{\mathcal{A}_i\}, P, R, \{\Omega_i\}, O, \gamma \rangle, \tag{10}$$

where $\mathcal{I}$ is the set of SUVs, $\mathcal{S}$ is the joint environment state, $\mathcal{A}_i$ and Ω_i are the local action and observation spaces of vehicle i, $P(s_{t+1} \mid s_t, \mathbf{a}_t)$ is the transition probability, $R(s_t, \mathbf{a}_t, s_{t+1}) \in \mathbb{R}^N$ is the local-reward vector, and $O(\mathbf{o}_{t+1} \mid s_{t+1}, \mathbf{a}_t)$ is the joint observation function. The shared team reward before clipping is $\bar{r}_t = N^{-1}\sum_{i=1}^N r_{i,t}$, where $r_{i,t}$ is the ith element of $\boldsymbol{R}(s_t, \mathbf{a}_t, s_{t+1})$, and γ is the discount factor.

Table 1
Environmental datasets used to construct the 500-TC observation environments. The dataset integrates ERA5 surface winds, HYCOM currents at seven depths, and IBTrACS best tracks.

Data product	Source	Variables used	Native grid	Native time
Surface wind at 10 m above sea level	ERA5	u_{10}, v_{10}, speed, direction	$0.25° \times 0.25°$	Hourly
Current at 0–60 m ocean depth	HYCOM	water_u, water_v, speed at 0/10/20/30/40/50/60 m	$\sim 0.04°$ lat $\times 0.08°$ lon	3-hourly
Best track	IBTrACS	Position, time, wind, pressure	Point track	Mainly 3/6-hourly

Note: although the SUV platform can carry sensors for observing temperature, salinity, and currents, the reinforcement-learning policy uses only surface wind and subsurface current observations as environmental inputs.

At each 10 min interval, the policy forms a closed perception–decision–action loop and does not follow a precomputed route:

$$\begin{aligned}\mathbf{o}_{i,t} &\longrightarrow \mathbf{a}_{i,t} = \pi_\theta(\mathbf{o}_{i,t}) \\ &\longrightarrow \mathbf{x}_{i,t+1} \longrightarrow \mathbf{o}_{i,t+1}.\end{aligned} \tag{11}$$

The updated observation becomes the next control input. This feedback structure lets the fleet adapt to changing wind and current fields, partial observability, and deployment uncertainty without a static waypoint sequence. Fig. 2 summarizes this interaction between the moving atmosphere–ocean environment and the cooperative SUV fleet.

3. Data and Simulation Environment

3.1. Reanalysis and Best-Track Products

ERA5 supplies hourly zonal and meridional wind components at 10 m above the sea surface on a regular 0.25° grid [1] [60]. Wind speed and direction are calculated from u_{10} and v_{10}. HYCOM supplies three-hourly zonal and meridional currents at ocean depths of 0, 10, 20, 30, 40, 50, and 60 m below the sea surface [2] [61]. The native horizontal grid is approximately 0.04° in latitude by 0.08° in longitude. IBTrACS provides TC time, center position, maximum wind, minimum pressure, and related metadata at mostly 3 or 6 h intervals [3] [62]. The dataset-selection domain used to identify candidate systems is 98.7°E–171.6°E and 6.8°N–56.7°N. Individual event fields are stored on individual TC crops, so a track may extend beyond this screening box. The overall event set spans 19 November 2007 to 3 December 2025 UTC. The surface observation variable is the 10 m wind speed and direction, whereas the subsurface observation variables are the horizontal current velocity, represented by water_u and water_v, and the derived current speed over the 0–60 m layer. Table 1 summarizes the products, variables, native grids, and temporal resolutions used to construct the simulation environments.

[1] ERA5 download portal: https://cds.climate.copernicus.eu/datasets/reanalysis-era5-single-levels.

[2] HYCOM GLBy0.08 dataset: https://www.hycom.org/data/glby0p08/expt-93pt0; HYCOM ESPC-D-V02 dataset: https://www.hycom.org/data/espcd-v02.

[3] IBTrACS download portal: https://www.ncei.noaa.gov/products/international-best-track-archive.

3.2. Spatiotemporal Alignment and Track Segmentation

Hourly ERA5 winds and three-hourly HYCOM currents are linearly interpolated to a common 10 min simulation clock and queried at each SUV position. The environmental trajectories are precomputed and do not respond to vehicle motion; this one-way forcing assumption is appropriate for planning at the scale of a TC. The policy receives only local reconstructed values at the current time, not future fields or the complete reanalysis map. Invalid interpolated values are replaced by a neutral value, and all policy observations are checked for finiteness. The best-track center is used only for training rewards and evaluation.

Commands are held constant within each decision interval:

$$\mathbf{a}_i(t) = \mathbf{a}_{i,k}, \quad t_k \le t < t_{k+1}, \quad t_{k+1} - t_k = \Delta t = 10\,\text{min}. \tag{12}$$

The environmental fields, TC track, action, and position update use the same causal time grid. The 10 min step balances control responsiveness against the hourly and three-hourly resolution of the source products; a shorter step would add samples without adding resolved information. The same interval is used for IPPO and all traditional planners.

Each IBTrACS track is first linearly interpolated to the 10 min simulation interval. The cumulative along-track distance is computed using the local equirectangular approximation

$$\Delta s_k = \sqrt{\left[111\Delta\lambda_k \cos(\varphi_k)\right]^2 + \left(111\Delta\varphi_k\right)^2}, \tag{13}$$

where $\Delta\lambda_k$ and $\Delta\varphi_k$ are longitudinal and latitudinal increments in degrees, and 111 km is the mean meridional arc length per degree. Target segment lengths are sampled uniformly from 300–500 km, with a fixed 250 km stride, 50–250 km overlap, and an average overlap of about 150 km. Residual track shorter than 300 km is discarded. After assignment to 10 min track samples, realized lengths range from 292.1 to 507.1 km, with a median of 396.4 km.

The 500-TC dataset yields 7,248 segments from 495 TCs. The 450-TC training split contributes 6,480 segments from 445 usable TCs, whereas the 50 held-out TCs contribute 768 segments. The split is made at the TC level, so no TC appears in both sets. Segment durations range from 2.2 to 103 h. The median duration was 16.5 h and

Table 2
Public Blue Whale reference and the two simulation configurations. The second simulation column is a simple parameterization of the present-day reference platform, whereas the third represents a possible future-upgrade configuration with higher surface and subsurface speeds for comparison. These scenarios are modeling assumptions, not certified vehicle data. The 20 h value is a common simulation horizon, not a platform-endurance claim.

Property	Public reference	Current-platform scenario	Future-upgrade scenario
Surface speed	23 kn, ≈11.83 m/s	12 m/s	20 m/s
Submerged speed	3 kn, ≈1.54 m/s	1.5 m/s	2.0 m/s
Public-reference submerged endurance	≥4 h at 3 kn; ≥72 h standby	–	–
Underwater evaluation horizon	–	20 h simulation	20 h simulation
Sampling depth	≈60 m	0–60 m, seven layers	0–60 m, seven layers
Payload	≥800 kg	800 kg	800 kg
Surface fleet size	Single platform	4/8/12/16	4/8/12/16
Decision interval	Not specified	10 min	10 min
Surface formation	–	100 km ring, $360°/N$	100 km ring, $360°/N$

the mean was 20.0 h. These segments provide mission-scale evaluation windows; battery depletion is not simulated. The platform can carry current, CTD, pH, chlorophyll, and turbidity sensors [28], while the control experiments use current velocity as the decision-making observation.

4. Multi-Agent Reinforcement Learning Method

4.1. Blue Whale-Class Platform Modeling

Each SUV is represented by a horizontal point mass model with mode-dependent speed bounds and position updates from Eqs. (1)–(2). The surface actor uses globally shared 10 m wind observations and its normalized velocity, whereas the underwater actor uses a causal five-frame history of its local current and normalized velocity. The surface control variable is defined in the Earth-fixed frame as a desired ground velocity. Therefore, ocean-current compensation is not included as a separate term in the surface position update. Underwater motion retains advection because 1.5–2.0 m/s propulsion is comparable to the TC-induced current.

The two simulation configurations compare present and potential future platform capabilities. The 12 m/s surface and 1.5 m/s submerged limits are close to the public Blue Whale design speeds, whereas 20 m/s and 2.0 m/s represent a future-upgrade case. Table 2 reports the complete parameterization and distinguishes modeling assumptions from certified specifications.

Across 32,352 IBTrACS track samples, the median, 95th-percentile, and 99th-percentile TC translation speeds are 5.14, 13.38, and 20.06 m/s (Table A.1). The 12 and 20 m/s surface limits cover 93.8% and 98.9% of these samples, respectively. The submerged limits cover only 3.7% and 7.8200 km initialization region and does not attempt to overtake and follow the TC center.

4.2. Surface and Subsurface Reward Design

The surface objective is to follow the translating TC as closely as possible and reach the high-wind region, with the final tracking error as the primary operational metric. Let $d_{i,k+1}$ denote the post-transition distance from SUV i to the TC center after executing the command at decision step k. The surface reward used in the implementation is

$$\begin{aligned} r_i^{\mathrm{surf}} = & -\frac{\min(d_{i,k+1}, D_{\max})}{D_{\mathrm{scale}}} + b_{\mathrm{near}} \mathbf{1}_{d_{i,k+1}<d_{\mathrm{near}}} \\ & - \frac{W_{\mathrm{form}} \max(0, d_{i,k+1} - R_{\mathrm{form}})}{D_{\mathrm{scale}}} \\ & - \lambda_u \|\bar{\mathbf{a}}_{i,k} - \bar{\mathbf{a}}_{i,k-1}\|_2^2 + r_i^{\mathrm{coast}}, \end{aligned} \tag{14}$$

where the first term is a bounded proximity cost and the second term provides a discrete near-center bonus with $b_{\mathrm{near}} = 1$. Here $D_{\max} = 300$ km caps the proximity cost, $D_{\mathrm{scale}} = 100$ km normalizes the distance terms, and $d_{\mathrm{near}} = 100$ km defines the near-center reward region. The third term adds an additional radius-excess penalty beyond $R_{\mathrm{form}} = 100$ km, and the fourth term penalizes abrupt action changes with $\lambda_u = 0.05$. The term r_i^{coast} represents the coastal-safety reward defined below. The 300 km cap prevents distant errors from producing an excessively steep gradient and reduces premature policy collapse during early training. After aggregation over SUVs, the team average reward is clipped to $[-5, 2]$.

For subsurface observation, four SUVs are assigned to each prescribed depth. Their primary objective is not to follow the surface TC center directly, but to observe the highest possible current speed. Geometric and current-weighted spatial coverage are then evaluated as complementary measures of observation extent and observation value. Low-speed SUVs cannot generally overtake a translating TC, and the vertical projection of the surface center is not necessarily

the location of the strongest subsurface current. For subsurface IPPO training, the current-seeking reward is

$$r_i^{\mathrm{sub}} = \min\left(\frac{\|\mathbf{u}_{i,k}\|}{V_{\mathrm{norm}}}, R_{\max}\right) - \lambda_a \|\tilde{\mathbf{a}}_{i,k}\|_2^2 + \beta_{\parallel} \max(0, \cos\theta_i) + r_i^{\mathrm{coast}}, \tag{15}$$

where $V_{\mathrm{norm}} = 1.0$ m/s, $R_{\max} = 2.0$, $\lambda_a = 0.05$, and $\beta_{\parallel} = 0.5$. The term $-\lambda_a\|\tilde{\mathbf{a}}_{i,k}\|_2^2$ penalizes large normalized propulsion commands and regularizes the action magnitude. It is not a calibrated battery-energy or power-consumption model. The angle θ_i denotes the direction difference between the effective propulsion velocity and the pre-move current $\mathbf{u}_{i,k}$, so the positive-alignment term is activated only when the SUV pushes downstream; formally, we define $\cos\theta_i = 0$ when either vector is zero and otherwise set $\cos\theta_i = (\mathbf{q}_{i,k}^{\mathrm{prop}} \cdot \mathbf{u}_{i,k})/(\|\mathbf{q}_{i,k}^{\mathrm{prop}}\|_2 \|\mathbf{u}_{i,k}\|_2)$. The current-intensity reward is evaluated at $\mathbf{x}_{i,k}$ before the state update.

4.3. Coastal Safety and Action Smoothing

The land mask is converted into a nonnegative offshore-distance field. A vehicle inside land receives a large penalty, and a vehicle within the buffer distance D_{buf} receives

$$r_i^{\mathrm{coast}} = \begin{cases} -P_{\mathrm{land}}, & D_i = 0, \\ -W_b \max(0, 1 - D_i/D_{\mathrm{buf}}), & 0 < D_i < D_{\mathrm{buf}}, \\ 0, & D_i \geq D_{\mathrm{buf}}, \end{cases} \tag{16}$$

where $D_i = D(\mathbf{x}_{i,k+1})$ is the post-transition offshore distance, with $D_i = 0$ on land, and P_{land} and W_b are positive penalty coefficients.

For the surface stage, three complementary mechanisms reduce coastline intrusion: offshore-only rejection sampling during deployment, a soft nearshore penalty, and an action-level safety filter. The main surface experiments use a minimum deployment distance of 100 km, a 100 km nearshore buffer, and a 100 km safety margin. Best-of-rejection deployment favors offshore initial states, the reward discourages trajectories near land, and the safety filter removes the onshore action component when an SUV approaches the shoreline. For the subsurface stage, the nearshore penalty is retained and an episode is truncated at landfall, but offshore-only initialization and action-level safety filtering are not applied. These mechanisms reduce land exposure in the surface results but do not guarantee offshore operation for every vehicle position.

To reduce oscillatory control, the main surface experiments use an exponential action filter:

$$\mathbf{a}_k^{\mathrm{filt}} = (1 - \beta)\mathbf{a}_k^{\mathrm{raw}} + \beta \mathbf{a}_{k-1}^{\mathrm{filt}}, \tag{17}$$

where $\beta \in [0, 1)$ controls the degree of temporal smoothing. The filter uses only the current and previous actions, which prevents future-information leakage. The filtered command $\mathbf{a}_k^{\mathrm{filt}}$ is then passed through the unit-disk projection and speed mapping in Eq. (9).

4.4. Parameter-Sharing Independent PPO

All SUVs contribute to the same observation and tracking objective, and the training return is defined as the team-average reward. This return defines a fully cooperative task. Its reward structure motivates the use of IPPO, in which decentralized vehicle actions are optimized against a common team objective while policy parameters are shared across vehicles.

In this paper, IPPO denotes the cooperative and parameter-sharing implementation of independent PPO. All SUVs use the same actor and critic weights, while their observations and action samples remain vehicle-specific. Actions are sampled independently, and the per-agent values are summed for training without a global-state centralized critic. The main surface model uses a memoryless multilayer perceptron, whereas the active underwater controller uses a five-frame current history. The shared cooperative return is the team-average reward, with value estimate $V_\phi(\mathbf{o}_t) = \sum_{i \in \mathcal{I}} v_\phi(\mathbf{o}_{i,t})$, and a single clipped PPO objective is optimized over the factorized multi-agent action vector:

$$L^{\mathrm{CLIP}}(\theta) = \mathbb{E}_t\Big[\min\Big(\rho_t(\theta)\hat{A}_t, \ \mathrm{clip}\left(\rho_t(\theta), 1 - \epsilon, 1 + \epsilon\right)\hat{A}_t\Big)\Big], \tag{18}$$

where

$$\rho_t(\theta) = \prod_{i \in \mathcal{I}} \frac{\pi_\theta(a_{i,t} \mid o_{i,t})}{\pi_{\theta_{\mathrm{old}}}(a_{i,t} \mid o_{i,t})}, \tag{19}$$

and $\hat{A}_t$ is estimated by generalized advantage estimation from the team-average reward. The actor objective is complemented by a clipped value loss and an entropy bonus. During execution, each SUV computes its action from its own observation and the shared policy network. The complete training and deployment procedure is summarized in Algorithm 1.

To compare reinforcement-learning algorithm choices, IPPO, SAC, and TD3 were trained for the same 2×10^6 environment steps on representative surface and subsurface cases. In the surface comparison, IPPO achieved a final error of 312.0 km, compared with 510.0 km for SAC and 526.2 km for TD3. In the subsurface comparison, IPPO obtained an episode-level current-weighted coverage C_w of 1532.7, compared with 1211.6 for SAC and 1239.8 for TD3. IPPO also required no replay buffer in these representative runs. We use IPPO for the main, formation, and planner comparisons and treat this algorithm comparison as bounded evidence without claiming a universal ranking.

The complete reward coefficients, safety parameters, network dimensions, and PPO hyperparameters are summarized in Appendix C.

4.5. Surface and Subsurface Evaluation

The center-distance reward is used only during training and is absent from validation. Validation tests whether a trained policy can infer the TC direction from wind observations in unseen events. The primary surface metric is the

Algorithm 1 Training and deployment of parameter-sharing IPPO

Input: Training TC split, stage-specific environment, fleet size N, speed limit $V_{\max}$, reward coefficients, and random seed
Output: Shared actor and critic parameters

Initialize the actor, critic, and parallel environment workers
for training iteration $m = 1, \dots, M$ **do**
  Reset the stage-specific environment and obtain $\mathbf{o}_{i,t}$ for all $i = 1, \dots, N$
  for rollout step t **do**
    Sample or select an action for each SUV from $\pi_\theta(\cdot \mid \mathbf{o}_{i,t})$
    Apply the stage-specific action filter, project each agent's action onto the unit disk, and apply the safety filter
    Advance all SUVs and collect the team-average reward
  end for
  Compute GAE advantages and returns for the team-average reward
  Update the shared actor and critic with the clipped PPO objective
end for
Freeze the actor and load the same parameters on every SUV
for each deployment episode **do**
  Observe local or shared data according to the stage communication model
  Compute $\mathbf{a}_{i,t}$ from the shared actor
  Apply action smoothing, circular projection, and safety filtering
  Execute the action and observe the next state
end for

final fleet-averaged SUV-to-TC-center distance. A smaller value places the fleet closer to the TC center, where near-core winds are generally stronger. Mean tracking error is reported as a secondary process diagnostic.

For training seed s and its validation dataset, the seed-level value is

$$D^{(s)}_{\text{final}} = \frac{1}{M_s} \sum_{e=1}^{M_s} \frac{1}{N} \sum_{i=1}^{N} \left\| \mathbf{x}_{i,e}(T_e) - \mathbf{x}_{\text{TC},e}(T_e) \right\|_2 , \quad (20)$$

where T_e is the final decision time of validation segment e, and M_s is the number of validation segments. Results are averaged over ten training seeds and reported with the sample standard deviation.

The process-level fleet-mean tracking error for the same seed is

$$D^{(s)}_{\text{avg}} = \frac{1}{M_s} \sum_{e=1}^{M_s} \frac{1}{K_e} \sum_{k=1}^{K_e} \frac{1}{N} \sum_{i=1}^{N} \left\| \mathbf{x}_{i,e,k} - \mathbf{x}_{\text{TC},e,k} \right\|_2 , \quad (21)$$

where K_e indexes the post-transition decision intervals. Distance is averaged over SUVs, episode time, and validation segments, then over training seeds.

Surface evaluation covers 4/8/12/16 SUVs on a 100 km ring with equal angular spacing. Training samples formation-center offsets within an 800 km radius; the trained policies are evaluated at 100, 200, 400, 600, and 800 km. These are mission-scale test conditions, not tuned deployment optima. Metrics are first averaged over validation segments within each seed, then summarized over ten seeds. Paired 12-versus-20 m/s tests are two-sided and uncorrected across the twenty fleet-size/radius comparisons, so their p-values are exploratory.

For the subsurface evaluation, each depth from 0, 10, 20, 30, 40, 50, and 60 m is tested as a separate 20 h configuration with four SUVs and 1.5 or 2.0 m/s speed limits. The 200 km terminal surface tracking error marks the assumed start of descent and is used as the upper bound of the subsurface initialization radius at the corresponding surface-segment end time. The descent time is neglected: each SUV is initialized directly at its assigned depth. The surface endpoint positions themselves are not transferred. There is no inter-vehicle communication, so each SUV uses only its own five-frame current history and propulsion state. IPPO and the traditional planners are evaluated on the same episodes, deployment seeds, motion model, and speed bound. The IPPO policy uses its deterministic mean action during evaluation.

For evaluation reporting, let K be the number of recorded position increments and let $k = 0, \dots, K$ index the archived positions. The mean experienced current speed and geometric coverage are

$$\bar{V}_i(z) = \frac{1}{K+1} \sum_{k=0}^{K} \left\| \mathbf{u}_{\text{ocean}}(\mathbf{x}_{i,k}, z, t_k) \right\|_2 . \quad (22)$$

$$C(z) = A_{\text{cell}} \left| \bigcup_{i=1}^{N} \bigcup_{k=0}^{K} \mathcal{G}(\mathbf{x}_{i,k}) \right| , \quad (23)$$

where $\mathcal{G}(\mathbf{x})$ is the 0.25° grid cell containing position $\mathbf{x}$, $|\cdot|$ counts unique cells, and $A_{\text{cell}} \approx (0.25° \times 111 \text{ km})^2$ is the nominal grid-cell area used for cross-method comparison. The figures report the fleet means $N^{-1} \sum_{i=1}^{N} \bar{V}_i(z)$ and the geometric coverage $C(z)$. Because neither current intensity nor coverage alone captures observation value, we also report the episode-level current-weighted coverage

$$C_{\text{w}}(z) = \frac{1}{M} \sum_{e=1}^{M} \left[\frac{1}{N} \sum_{i=1}^{N} \bar{V}_{i,e}(z) \right] C_e(z), \quad (24)$$

where M is the number of paired validation episodes. This quantity assigns a higher value to covering a broad region while encountering a strong current within that region, without assuming that either objective dominates in every episode.

4.6. Traditional Planning Baselines

We include four non-learned planners as secondary references [36, 38, 40]. The artificial-potential-field planner moves along the local gradient of observed wind or current speed and adds an offshore repulsion inside the coastal buffer. The receding-horizon A* planner builds an online local value map, re-plans a three-step minimum-cost route, executes the first action, and refreshes the plan every 15 decisions. The boustrophedon lawnmower planner follows fixed parallel coverage legs with phase offsets between vehicles and uses no environmental value map. The information-greedy planner evaluates eight candidate unit actions and selects the one with the largest predicted local wind or current value plus the coastal preference.

None of the planners receives the TC center, the complete reanalysis field, future frames, or the learned policy state. Surface planners use the same shared wind samples and fleet navigation state as IPPO. Underwater, each vehicle maintains a private map because communication is disabled. All four use the same 10 min interval, speed envelope, circular projection, surface safety filter, and subsurface landfall termination. They require no training and have no training-seed variance. Surface planners are evaluated on all 768 held-out segments. Underwater planners use the same 500 paired episodes per depth-speed setting. These episodes comprise 50 test TCs with ten deployment seeds each.

5. Simulation and Results

5.1. Surface Active Tracking

Using the evaluation protocol in Section 4.5, we evaluate the trained surface policies across four fleet sizes, two speed limits, five deployment radii, and ten independent training seeds. Final tracking error in Eq. (20) is the primary metric. Mean tracking error is a secondary diagnostic of how closely the fleet remains to the moving TC, while path length and mean ground speed characterize motion demand.

Figure 4 reports the population-level speed-limit comparison. Averaged over fleet sizes and training seeds, the 20 m/s policy obtains final errors of 204.8, 217.2, 272.5, 379.3, and 540.8 km at deployment radii of 100, 200, 400, 600, and 800 km. The corresponding 12 m/s values are 236.7, 255.1, 335.5, 468.4, and 651.3 km. Thus, 20 m/s gives the lower final error in all 20 matched fleet-size/radius comparisons, and its fleet-averaged advantage grows from 31.9 km at 100 km to 110.5 km at 800 km. Both policies remain farther from the center as deployment error increases, because finite speed and episode duration cannot remove every initial displacement.

The mean tracking error shows the same ordering: 20 m/s has the smaller distance in all 20 matched comparisons, with a median 12-minus-20 m/s paired difference of 49.7 km. The two-sided paired tests give $p < 0.05$ in every case; these tests are exploratory because they are not corrected across the 20 comparisons. The final-error benefit is larger than the mean tracking-error benefit at both ends of the deployment range, indicating that the higher-speed policy continues reducing residual separation late in the episode.

The 12 m/s policy reduces path length per SUV by 83.4–220.1 km across the five radii and lowers mean ground speed by 1.75–3.93 m/s, whereas the 20 m/s policy achieves closer final tracking.

The representative trajectories clarify how the aggregate result is produced. Fig. 3 follows the four-SUV, 400 km KHANUN case. The vehicles first turn along the observed wind structure and then reduce their separation from the center; by 19.5 h they have entered the high-wind branch. Individual trajectories differ because each SUV has its own state, but the fleet-level motion follows a consistent inward correction.

Figure A.1 separates the effects of fleet size and deployment error. At 100–200 km, the formations converge rapidly and stay concentrated near the TC core. At 600–800 km, trajectories lengthen and fan out before turning inward, and final positions remain farther from the center. Increasing the fleet from 4 to 16 SUVs adds sample branches and broadens the approach region, but it does not replace the long-range transit. In the representative 12- and 16-SUV cases, paths overlap near the core once the shared wind observations guide the fleet toward the same high-wind region. Deployment error therefore controls the required transit distance, whereas fleet size mainly controls how broadly the approach is sampled.

5.2. Surface Formation Comparison

The ring, line, grid, and V-shaped formations provide four different ways to allocate spatial observations across eight SUVs. The V formation places four SUVs on each arm, with the two arms separated by 70° and extending from 25 to 100 km along the arm. Each policy was trained with the same 2-million-step protocol, deployment distribution, and safety layer at both 12 and 20 m/s, then evaluated without retraining at 100–800 km (Table A.2). Averaged over the ten speed–radius conditions, the line gives the lowest mean final error of 320.8 km, followed by the V formation at 323.1 km, the ring at 341.2 km, and the grid at 396.0 km. The line is best in six speed–radius conditions, and the V formation is best in the remaining four conditions at 100–600 km under 12 m/s. The ring is best at 800 km under 12 m/s. The V formation also lowers the mean tracking error to 330.8 km, compared with 332.5 km for the line, 348.4 km for the ring, and 377.4 km for the grid.

Formation geometry is summarized at each post-transition state $\mathbf{x}_{i,e,k}$. With centroid $\bar{\mathbf{x}}_{e,k} = N^{-1}\sum_i \mathbf{x}_{i,e,k}$, the episode means are

$$R_{\mathrm{form},e} = \frac{1}{K_e}\sum_{k=1}^{K_e}\frac{1}{N}\sum_{i=1}^{N}\left\|\mathbf{x}_{i,e,k} - \bar{\mathbf{x}}_{e,k}\right\|_2, \tag{25}$$

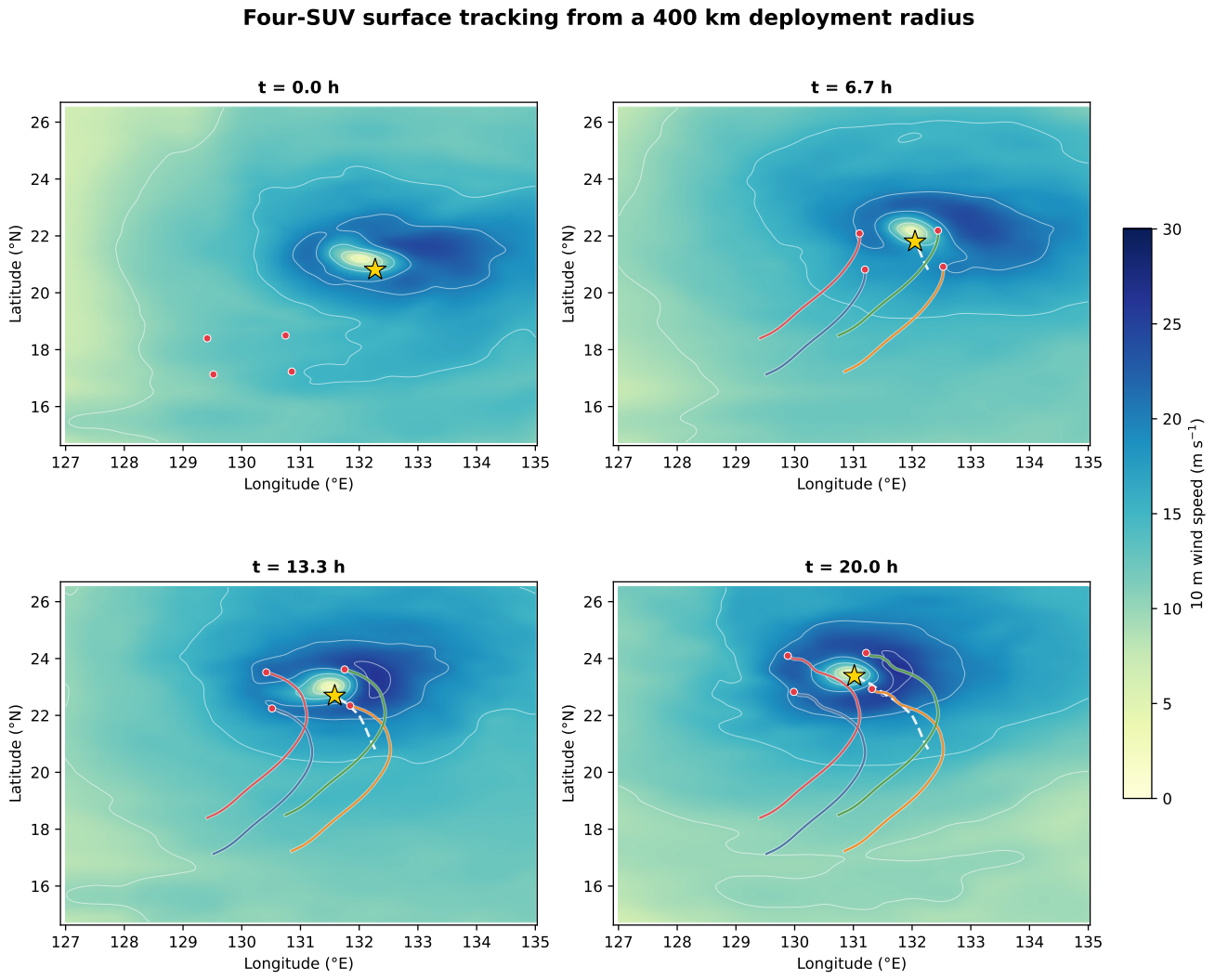


Figure 3: Four-SUV surface-tracking process for KHANUN (2023) at the 20 m/s speed limit. The four panels show the initial deployment and subsequent states at approximately 6.5, 13.0, and 19.5 h. The color background is the spatially interpolated ERA5 10 m wind-speed field at each time. Small red dots mark vehicle positions, and the yellow star marks the TC center.

$$S_{\mathrm{pair},e} = \frac{1}{K_e} \sum_{k=1}^{K_e} \frac{2}{N(N-1)} \sum_{1 \le i < j \le N} \left\| \mathbf{x}_{i,e,k} - \mathbf{x}_{j,e,k} \right\|_2 . \quad (26)$$

Reported formation radii and pairwise spacings are means of $R_{\mathrm{form},e}$ and $S_{\mathrm{pair},e}$ over the 768 validation segments at each speed–radius condition.

The grid produces the shortest per-vehicle path, averaging 392.7 km, followed by the ring at 416.5 km, the line at 482.6 km, and the V formation at 499.8 km. The V formation gives the most compact geometry, with a mean formation radius of 44.4 km and pairwise spacing of 63.9 km. The line gives a 57.9 km radius and 86.3 km spacing. The ring retains a 99.3 km radius and 142.6 km spacing, whereas the grid preserves the widest footprint, with a 103.1 km radius and 156.1 km spacing and the largest final tracking error. The compact V geometry therefore provides the closest tracking under 12 m/s, but its longer paths show that this benefit comes with greater transit effort.

Figure 5 visualizes one matched KHANUN segment. The formation geometry is not rigidly preserved because each policy moves under the same shared wind observation but with its own state. This representative episode illustrates how the compact V and line formations remain spatially concentrated, whereas the ring and grid retain a broader sampling footprint. The choice of formation is based on the aggregate 768-segment results, not on this episode. We use the ring for the main experiments. The V and line formations track more closely in several conditions, and the grid gives shorter paths with wider spacing. The alternatives are reported as sensitivity cases; no formation is universally best.

5.3. Execution-Layer Action Filtering

In preliminary rollouts, the unfiltered commands changed direction frequently, causing the SUVs to turn repeatedly and accumulate avoidable path length and

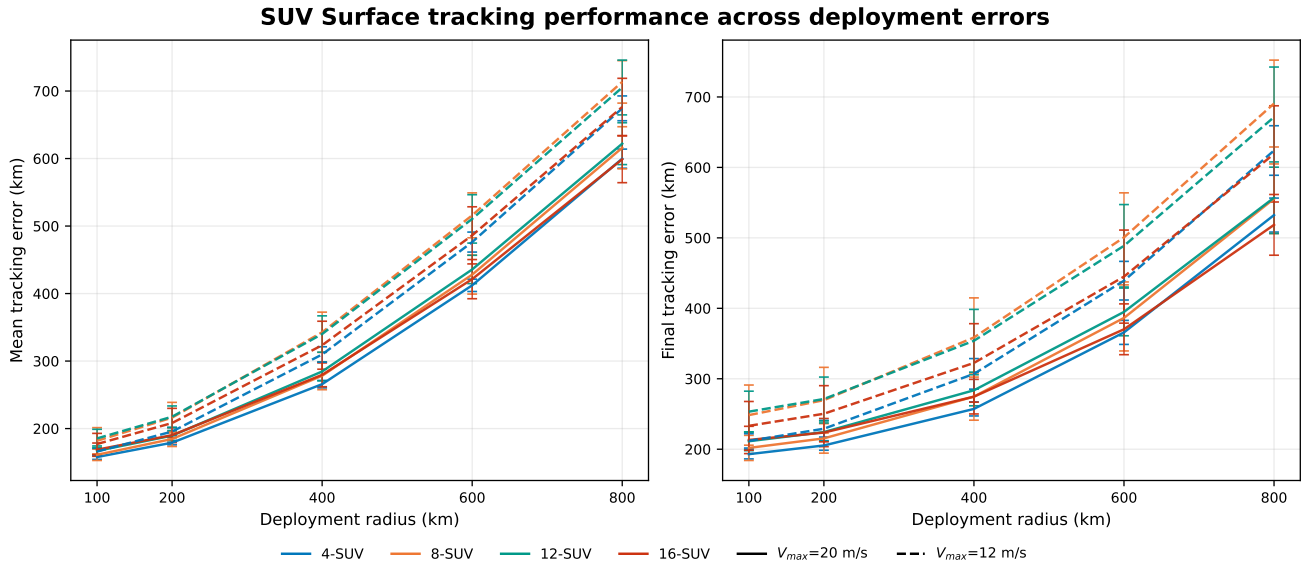


Figure 4: SUV surface tracking performance across deployment errors. The left panel shows the fleet-mean tracking error to the TC center, representing whether the fleet remains close to the moving TC throughout an episode. The right panel shows the final tracking error to the TC center at the end of the episode; smaller values indicate a closer final position relative to the TC center. Colors denote fleet sizes and line styles denote the 20 and 12 m/s speed limits. Each point summarizes ten independent training runs using ten random seeds. Error bars show the sample standard deviation.

propulsion changes. Fig. 6 illustrates this behavior: the unfiltered trajectories form sawtooth-like detours, whereas filtering produces smoother paths under the same policy and initial positions. We therefore added an execution-layer low-pass filter that smooths successive commands while preserving the 10 min decision interval. Table 3 quantifies this change using the same trained actors over identical validation episodes. Without filtering, the 4-, 8-, 12-, and 16-SUV configurations travel 1019.2–1160.0 km per vehicle, with mean speeds of 14.85–16.45 m/s and maximum speeds of 19.46–19.47 m/s. With the execution-layer filter ($\beta = 0.8$), path length falls to 497.4–670.8 km, mean speed to 8.49–11.01 m/s, and maximum speed to 14.25–17.97 m/s. The final tracking error changes from 355.4 to 251.3 km for four SUVs, from 324.7 to 277.8 km for eight SUVs, from 312.0 to 278.8 km for 12 SUVs, and from 260.2 to 256.1 km for 16 SUVs. Averaged over the four fleet sizes, it decreases from 313.1 to 266.0 km, a reduction of 47.1 km or 15.0%. Filtering improves final tracking for every fleet size.

The same comparison shows reductions for every fleet size: path length decreases by 651.7, 614.9, 521.9, and 459.1 km for 4, 8, 12, and 16 SUVs, respectively, while maximum speed decreases by 5.22, 4.14, 3.50, and 1.49 m/s.

5.4. Subsurface Observation and Comparison with Traditional Planners

Before comparing IPPO with non-learned planners, we first isolate its subsurface performance at each observation depth and propulsion limit. Appendix Table A.3 reports ten-seed mean experienced current speed, geometric coverage, and current-weighted coverage.

Mean experienced current decreases from 0.534 m/s at the surface to approximately 0.30 m/s at 60 m under both speed limits. At a fixed depth, the 1.5 and 2.0 m/s policies differ by less than 0.006 m/s in mean experienced current. The faster envelope mainly expands the sampled region: relative to 1.5 m/s, geometric coverage is 1.18–1.34 times larger and current-weighted coverage is 1.15–1.30 times larger.

Appendix Figure B.1 shows one matched trajectory example. Table 4 compares IPPO with four traditional planners. Its surface panel provides matched context, whereas the subsurface panel is the focus of this section. Across all 40 surface configurations, IPPO has the lowest final tracking error, mean tracking error, and path length. It produces a lower final error than potential field in 35 of 40 comparisons and than each remaining planner in all 40. Its path is shorter than those of potential field and A* in every comparison and shorter than those of the lawnmower and information-greedy planners in 26 and 34 of 40 comparisons. The path advantage over the latter two planners is therefore less uniform than the final-error advantage.

For the subsurface task, no planner leads all three metrics. Potential field has the highest mean experienced current at 0.452 m/s. IPPO has the largest geometric coverage at 3506 km^2 and the largest current-weighted coverage $C_w(z)$. Its mean current of 0.408 m/s is close to that of the lawnmower planner. IPPO's current-weighted coverage exceeds that of potential field, A*, lawnmower control, and information-greedy control in all 14 depth–speed settings by factors of 1.41–1.60, 1.39–1.56, 1.17–1.31, and 1.95–2.28, respectively. Thus, potential field performs better on mean experienced current, whereas IPPO performs better on

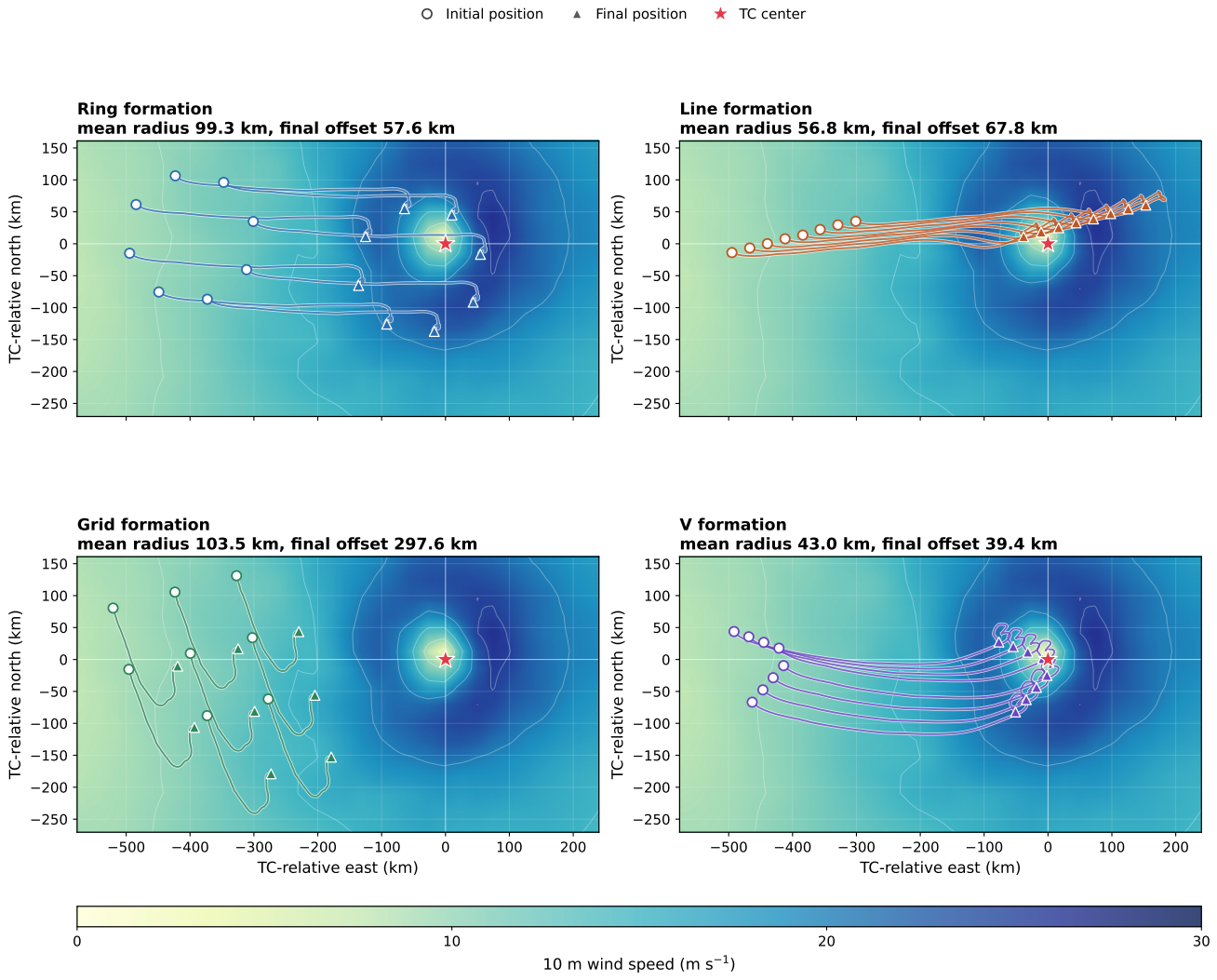


Figure 5: Representative surface trajectory comparison for ring, line, grid, and V-shaped formations with eight SUVs at 12 m/s. Open circles denote initial positions, triangles denote final positions, and the red star denotes the TC center. The color background and contours show the spatially interpolated ERA5 10 m wind-speed field at the final simulation step. Axes are relative to the instantaneous TC center.

Table 3
Execution-layer low-pass filtering for 4/8/12/16 surface SUVs on 768 held-out segments from 50 unseen TCs in the 500-TC dataset. Each trained actor is evaluated with the filter off and on ($\beta = 0.8$) on the same validation episodes.

Fleet	Filter	Final tracking error (km)	Path per SUV (km)	Mean speed (m/s)	Maximum speed (m/s)
4	Off	355.4	1160.0	16.45	19.47
4	On	**251.3**	**508.2**	**8.49**	**14.25**
8	Off	324.7	1119.5	16.11	19.47
8	On	**277.8**	**504.7**	**8.65**	**15.32**
12	Off	312.0	1019.2	14.85	19.46
12	On	**278.8**	**497.4**	**8.52**	**15.96**
16	Off	260.2	1129.9	16.29	19.47
16	On	**256.1**	**670.8**	**11.01**	**17.97**

Final tracking error is the fleet average of the individual SUV-to-best-track-center distances at the final time step. Off and on use the same trained actor.

geometric and current-weighted coverage; no method leads all three metrics.

Appendix Table B.1 confirms the same pattern after stratification. IPPO retains the lowest final error and mean tracking error at every surface deployment radius. Underwater, potential field has the highest mean current at every depth and IPPO has the largest geometric coverage.

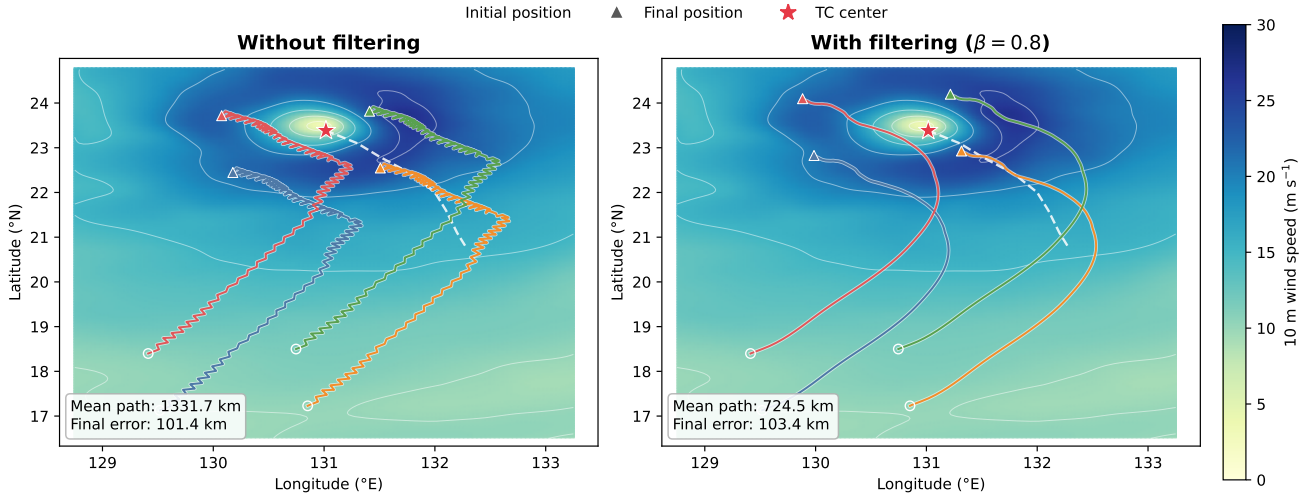


Figure 6: Representative effect of execution-layer filtering on surface paths for four SUVs under the same trained policy, initial positions, deployment radius, and KHANUN (2023) segment. The left panel uses the raw policy command, whereas the right panel applies the low-pass filter with $\beta = 0.8$. Open circles and triangles mark initial and final positions, respectively, and the red star marks the TC center.

Table 4
Comparison with four traditional planning baselines on the held-out TC segments from the same 500-TC train/test split. Surface values are means over the 4/8/12/16-SUV fleets, both speed limits, and all five deployment radii. Subsurface values are means over all seven depths and both propulsion limits. For surface tracking, lower errors and paths are better. For subsurface observation, higher current and coverage are better.

Panel A: surface tracking			
Method	Final error (km)	Mean tracking error (km)	Path per SUV (km)
Parameter-sharing IPPO	**356.2**	**356.6**	**462.3**
Potential field	413.4	404.2	728.8
Receding-horizon A*	457.2	429.4	823.9
Lawnmower coverage	554.7	478.4	509.1
Information greedy	466.2	429.4	528.7
Panel B: subsurface observation			
Method	Mean current (m/s)	Geometric coverage (km^2)	Current-weighted coverage (10^3 m s^{-1} km^2)
Parameter-sharing IPPO	0.408	**3506**	**1.599**
Potential field	**0.452**	2048	1.050
Receding-horizon A*	0.427	2310	1.097
Lawnmower coverage	0.408	2914	1.286
Information greedy	0.429	1549	0.739

6. Conclusion

This study developed and evaluated a parameter-sharing IPPO policy for coordinated surface–subsurface tropical-cyclone observation with Blue Whale-class SUVs. Current and future-upgrade speed settings were evaluated separately for the surface and subsurface fixed-depth tasks. Historical atmosphere–ocean reanalysis data, including ERA5 and HYCOM, were used to construct the simulation environments for SUV reinforcement-learning training. Training used 450 TCs, and evaluation used 50 unseen TCs.

At the surface, globally shared wind observations allowed the fleet to approach the moving TC despite uncertain deployment. The 20 m/s maximum-speed configuration produced smaller final and mean tracking errors at every deployment radius, and its final-error advantage grew with deployment error. The 12 m/s maximum-speed configuration produced shorter paths and lower speeds. Action filtering reduced mean final tracking error at every fleet size and shortened paths and peak speed. In the eight-SUV formation comparison, the line formation had the lowest mean final error, the grid had the shortest paths and widest spacing, and the ring provided a broad sampling footprint and was retained as the main configuration because no formation was uniformly best. In the matched surface comparison with four traditional planners, IPPO had the lowest final and mean tracking errors and generally shorter paths. It had a lower

final error than potential field in most matched cases and consistently lower error than the other three planners.

At the subsurface, four SUVs sampled one of seven fixed layers using only their local current histories. The comparison with four traditional planners showed no single winner across all underwater metrics: potential field obtained the highest mean current, while IPPO obtained the largest geometric and current-weighted coverage. Under the same 20 h horizon, increasing the submerged speed limit from 1.5 to 2.0 m/s expanded the sampled area at every depth. A faster future SUV offers greater spatial sampling capacity, provided that endurance and control stability remain feasible.

Nevertheless, several limitations remain. The SUV dynamics are represented by a simplified horizontal point mass model, and calibrated six-degree-of-freedom dynamics are not included. The study targets autonomous environmental sensing and path planning, whereas detailed low-level vehicle control is not modeled. The main experiments were trained and validated in simulation, and real-vehicle field experiments were not conducted. Future work can incorporate six-degree-of-freedom dynamics and low-level control and validate the algorithms through field experiments in real environments. It should also address the sim-to-real gap in reinforcement learning. In addition, TC wind speeds used as observation inputs to the surface policy may be underestimated in reanalysis datasets, which can affect policy decisions. Future work can use observed TC wind speeds to fine-tune the existing policy through transfer learning.

Code and package availability

The code and processed results are available at `https://github.com/DYHAI/SUV-MARL-TC`. The experiments used Python 3.11, PyTorch 2.11.0 (CUDA 12.8 build), Stable-Baselines3 2.9.0, and Gymnasium 1.2.2 for policy training; NumPy 2.4.6 and pandas 2.3.3 for numerical and tabular processing; SciPy 1.17.1, xarray 2026.7.0, netCDF4 1.7.4, and h5py 3.16.0 for analysis and NetCDF processing; and Matplotlib 3.11.1 for visualization.

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A. Supplementary Observation Results

Table A.1: IBTrACS TC translation speeds for the 500-TC dataset. The percentages denote the fraction of main-track samples not exceeding each platform speed. These values describe the speed envelope only; tracking also depends on bearing error and remaining segment duration.

Quantity	P50	P90	P95	P99	Samples	TCs
TC translation speed (m s^{-1})	5.14	9.77	13.38	20.06	32,352	500
Envelope	Samples within speed		Samples above speed		Interpretation	
1.5 m/s submerged	3.7%		96.3%		local sampling, not TC chasing	
2.0 m/s submerged	7.8%		92.2%		local sampling, not TC chasing	
12 m/s surface	93.8%		6.2%		current-platform scenario	
20 m/s surface	98.9%		1.1%		future-upgrade scenario	

Table A.2: Surface initial-formation comparison for eight SUVs. Each policy was trained with formation-center offsets sampled inside an 800 km radius and evaluated without retraining at fixed offsets of 100, 200, 400, 600, and 800 km on the same 768 held-out TC segments. Error is the final SUV-to-best-track-center distance and path is the per-vehicle trajectory length. Bold entries are the best value in each speed/radius row; error and path are assessed separately. All distances are in km.

Speed	Train radius	Eval radius	Ring		Line		Grid		V	
(m/s)	(km)	(km)	Error	Path	Error	Path	Error	Path	Error	Path
12	800	100	210.4	284.4	184.8	330.4	295.8	**219.3**	**165.0**	366.2
12	800	200	228.6	317.2	202.4	364.5	317.7	**241.2**	**183.9**	406.9
12	800	400	312.0	364.2	295.7	409.9	408.5	**272.8**	**284.6**	463.9
12	800	600	445.9	383.9	452.6	408.5	542.6	**291.4**	**438.1**	486.9
12	800	800	**636.5**	372.4	655.6	394.1	733.5	**285.9**	637.5	487.9
20	800	100	197.5	**356.2**	**156.4**	435.3	204.7	383.2	158.1	416.6
20	800	200	209.9	**404.6**	**168.0**	489.5	217.7	433.7	173.8	470.6
20	800	400	266.6	**502.5**	**228.4**	599.5	277.7	537.7	251.8	571.0
20	800	600	371.1	**571.9**	**342.1**	676.6	390.6	616.4	385.3	644.7
20	800	800	533.9	**607.7**	**522.0**	717.2	571.1	645.5	553.3	682.8

Table A.3: IPPO subsurface observation performance at seven depths and two propulsion limits. Values are ten-seed means over the held-out TC segments. Current-weighted coverage is reported in 10^3 m s^{-1} km^2.

Depth (m)	V_{max} (m/s)	Mean current (m/s)	Geometric coverage (km^2)	Current-weighted coverage
0	1.5	0.534	3376	2.012
0	2.0	0.534	3968	2.319
10	1.5	0.472	3385	1.744
10	2.0	0.476	4073	2.107
20	1.5	0.444	3402	1.660
20	2.0	0.445	4037	1.956
30	1.5	0.409	3253	1.481
30	2.0	0.408	3895	1.750
40	1.5	0.364	2851	1.191
40	2.0	0.368	3808	1.555
50	1.5	0.324	2907	1.069
50	2.0	0.328	3749	1.374
60	1.5	0.300	2818	0.959
60	2.0	0.305	3559	1.207

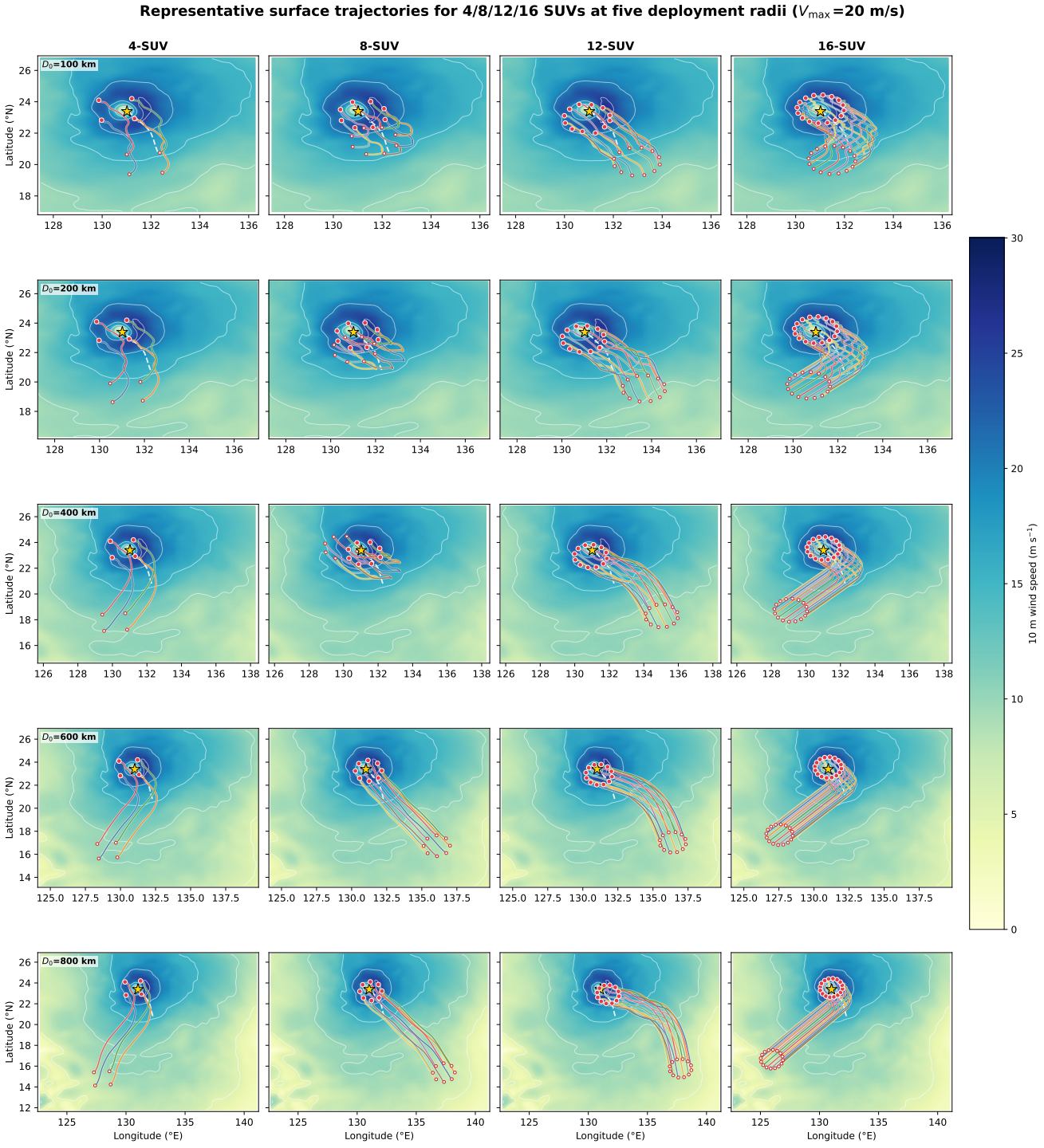


Figure A.1: Representative KHANUN (2023) trajectories for the final 20 m/s surface policy. Columns show 4-, 8-, 12-, and 16-SUV formations, while rows show nominal formation-center deployment radii of 100, 200, 400, 600, and 800 km. The color background is the spatially interpolated final-time ERA5 10 m wind-speed field. Thin colored lines are vehicle trajectories, open circles mark initial positions, small red dots mark final vehicle positions, and the yellow star marks the TC center at the end of the episode.

B. Traditional Planner Comparison

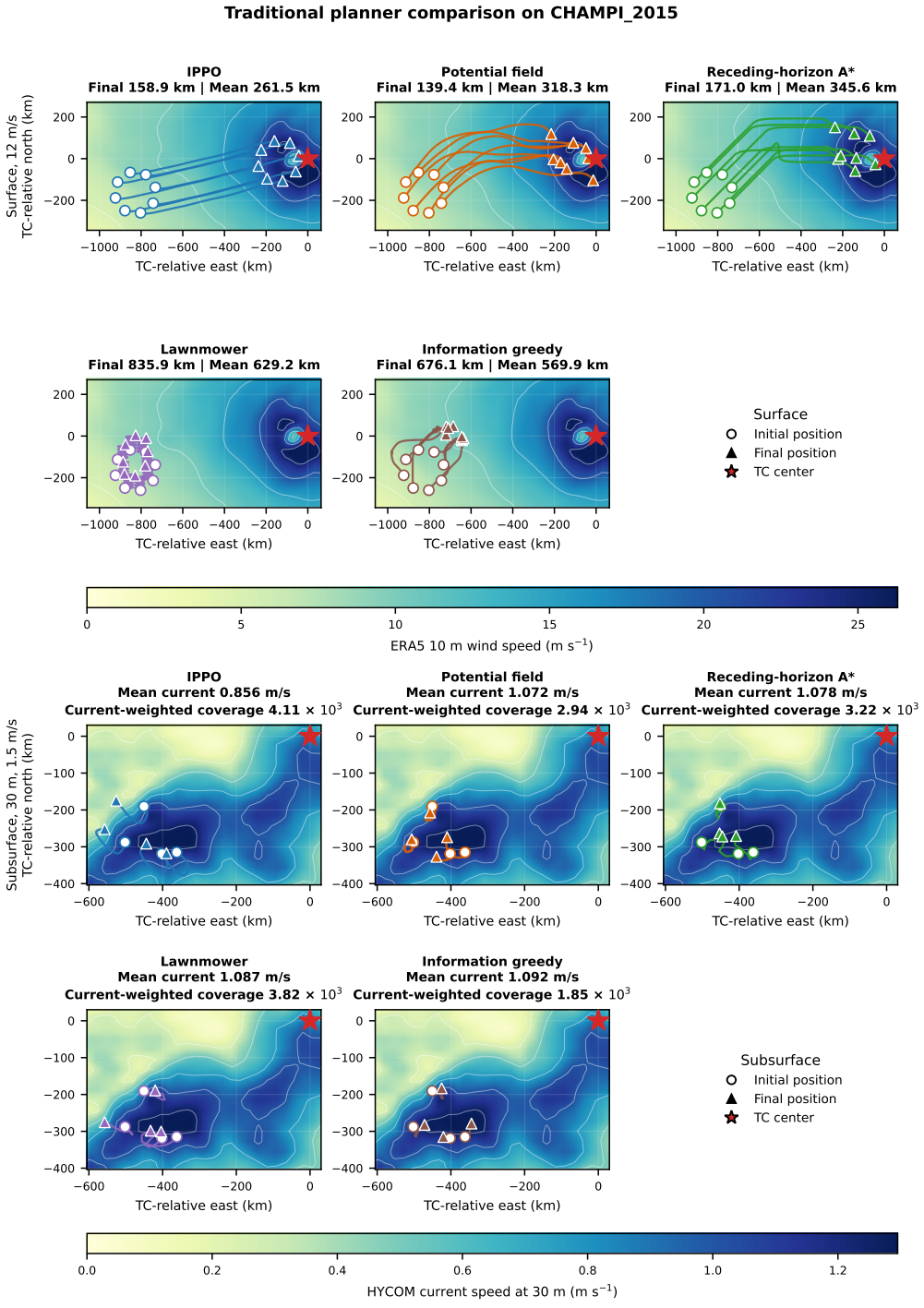


Figure B.1: Comparison between IPPO and the four traditional planners on the same CHAMPI segment. The upper two rows show 12 m/s surface trajectories over the ERA5 10 m wind-speed field. The lower two rows show 1.5 m/s subsurface trajectories at 30 m over the HYCOM current-speed field. All methods use the same initial deployment. Panel titles report final and mean tracking error for the surface task and mean experienced current for the subsurface task. Open circles mark initial positions, triangles mark final positions, and the red star marks the TC center.

Table B.1: Distance and observation metrics for the traditional planner comparison on the held-out TC segments. Panel A reports the two surface distance metrics by deployment radius, averaged over the four fleet sizes and both speed limits. Panel B reports mean experienced current speed, geometric coverage, and current-weighted coverage at each depth, averaged over the two propulsion limits. Current-weighted coverage is reported in 10^3 m s^{-1} km^2. For surface tracking, lower errors and paths are better. For subsurface observation, higher current and coverage are better.

Panel A: surface distance metrics						
Radius	Metric	IPPO	PF	A*	Lawn	Greedy
100 km	Final error (km)	**220.8**	290.4	325.8	366.1	295.2
	Mean tracking error (km)	**170.6**	229.9	250.5	232.9	212.0
200 km	Final error (km)	**236.1**	295.4	330.0	394.5	310.7
	Mean tracking error (km)	**197.6**	248.5	267.1	285.9	244.5
400 km	Final error (km)	**304.0**	353.5	394.9	504.5	405.1
	Mean tracking error (km)	**303.0**	343.0	367.4	441.8	374.5
600 km	Final error (km)	**423.9**	476.1	527.6	661.2	560.1
	Mean tracking error (km)	**460.8**	505.1	536.7	621.7	559.1
800 km	Final error (km)	**596.1**	651.8	707.6	847.4	759.8
	Mean tracking error (km)	**650.8**	694.4	725.0	809.7	756.7
Panel B: subsurface metrics by depth						
Depth	Metric	IPPO	PF	A*	Lawn	Greedy
0 m	Mean experienced current (m/s)	0.534	**0.577**	0.548	0.536	0.551
	Geometric coverage (km^2)	**3672**	2214	2445	3067	1579
	Current-weighted coverage	**2.335**	1.443	1.467	1.756	0.956
10 m	Mean experienced current (m/s)	0.474	**0.515**	0.489	0.472	0.492
	Geometric coverage (km^2)	**3729**	2189	2414	3020	1620
	Current-weighted coverage	**1.955**	1.249	1.284	1.513	0.876
20 m	Mean experienced current (m/s)	0.444	**0.486**	0.462	0.444	0.464
	Geometric coverage (km^2)	**3720**	2152	2388	2987	1632
	Current-weighted coverage	**1.877**	1.153	1.204	1.415	0.835
30 m	Mean experienced current (m/s)	0.409	**0.452**	0.429	0.410	0.431
	Geometric coverage (km^2)	**3574**	2105	2329	2939	1588
	Current-weighted coverage	**1.624**	1.039	1.104	1.291	0.754
40 m	Mean experienced current (m/s)	0.366	**0.412**	0.386	0.368	0.391
	Geometric coverage (km^2)	**3330**	1999	2273	2876	1542
	Current-weighted coverage	**1.469**	0.894	0.970	1.141	0.666
50 m	Mean experienced current (m/s)	0.326	**0.374**	0.349	0.327	0.351
	Geometric coverage (km^2)	**3328**	1876	2200	2803	1470
	Current-weighted coverage	**1.246**	0.750	0.848	0.986	0.575
60 m	Mean experienced current (m/s)	0.302	**0.350**	0.325	0.301	0.324
	Geometric coverage (km^2)	**3189**	1799	2120	2702	1410
	Current-weighted coverage	**1.092**	0.658	0.765	0.877	0.519

C. IPPO Algorithm Parameters

Table C.1 summarizes the reward, action-filter, and safety parameters used in the surface and underwater experiments. Table C.2 lists the shared IPPO network and training hyperparameters.

Table C.1: Surface and subsurface reward, action-filter, and coastal-safety parameters used in the final experiments.

Parameter	Symbol	Value	Description and applicable task
Panel A: surface reward and controller parameters			
Proximity-cost distance cap	$D_{\max}$	300 km	Distance cost is capped at 300 km to prevent excessively large gradients from distant initial deployments.
Distance normalization scale	D_{scale}	100 km	Normalization scale used by the proximity and radius-excess reward terms.
Near-center threshold	d_{near}	100 km	Threshold defining the discrete near-center bonus.
Near-center bonus	b_{near}	1.0	Bonus assigned when an SUV is within 100 km of the TC center.
Radius-excess threshold	R_{form}	100 km	Reference radius used by the additional distance penalty beyond 100 km.
Radius-excess weight	W_{form}	1.0	Weight of the penalty applied when an individual SUV is farther than 100 km from the TC center.
Action-change weight	λ_u	0.05	Weight of the squared action change $\|\bar{\mathbf{a}}_{i,k} - \bar{\mathbf{a}}_{i,k-1}\|_2^2$ between successive executed commands.
Action-filter coefficient	β	0.8	Coefficient of $\mathbf{a}_k^{\text{filt}} = (1-\beta)\mathbf{a}_k^{\text{raw}} + \beta\mathbf{a}_{k-1}^{\text{filt}}$. This is a controller setting and is not included in the reward.
Surface reward clipping range	$[r_{\min}, r_{\max}]$	$[-5, 2]$	Clipping range applied to the team-averaged surface reward.
Panel B: underwater IPPO parameters			
Current-speed normalization	V_{norm}	1.0 m/s	Current-intensity reward is implemented as $\min(\|\mathbf{u}\|/V_{\text{norm}}, R_{\max})$.
Current-reward cap	$R_{\max}$	2.0	Upper bound of the normalized current-speed reward.
Downstream-alignment weight	$\beta_{\parallel}$	0.5	Weight of the positive propulsion–current alignment term.
Normalized action-magnitude weight	λ_a	0.05	Penalizes the squared magnitude of the normalized propulsion command. This term regularizes the controller response and is not a calibrated energy or power model.
Underwater reward clipping range	$[r_{\min}, r_{\max}]$	$[-5, 5]$	Clipping range applied to the team-averaged underwater reward during IPPO training.
Panel C: coastal-safety parameters by stage			
Land penalty	P_{land}	20	Penalty applied when an SUV enters a land cell in either stage.
Nearshore buffer distance	D_{buf}	100 km	Width of the soft nearshore penalty buffer in both stages.
Nearshore penalty weight	W_b	8.0	Maximum soft-penalty weight within the 100 km nearshore buffer in either stage.
Minimum deployment offshore distance	D_{deploy}	100 km	Minimum coastline distance required by offshore-only rejection sampling during surface deployment; it is not applied to subsurface initialization.
Action-filter safety margin	D_{filter}	100 km	Surface-stage distance below which the onshore component of the commanded action is removed. The subsurface comparison does not use this filter.

Table C.2: Parameter-sharing IPPO network, training, and deployment hyperparameters used in the final surface and subsurface experiments.

Parameter	Symbol	Value	Description
Algorithm	–	PPO	Stable-Baselines3 PPO with parameter sharing and independent per-agent action sampling.
Return aggregation	–	Team average	The environment reward is averaged across the N SUVs before computing the shared PPO return.
Optimizer	–	Adam	Adam optimizer used for the actor and critic parameters.
Policy network	–	2×256 MLP	Actor: Linear(d_{obs}, 256)–Tanh– Linear(256, 256)–Tanh– Linear(256, d_{act}).
Value network	–	2×256 MLP	Critic: Linear(d_{obs}, 256)–Tanh– Linear(256, 256)–Tanh– Linear(256, 1). Per-agent values are summed during training.
Per-agent observation dimension	d_{obs}	16/28/40/52; 19	Surface observations for 4/8/12/16 SUVs are 16/28/40/52, respectively. The four-SUV underwater IPPO evaluation uses $d_{\text{obs}} = 19$.
Surface observation	–	Wind at 10 m above sea level	Using ideal surface communication, each SUV receives the current wind speed and components at 10 m above the sea surface observed by every SUV in the fleet, together with its normalized ground velocity and $[\cos\psi, \sin\psi]$.
Underwater observation	–	Current only	With no inter-vehicle communication, the active IPPO policy observes only its own local current speed and components at the assigned ocean depth, together with a five-frame history, its own normalized propulsion velocity, and $[\cos\psi, \sin\psi]$.
Action dimension	d_{act}	2 per SUV	Continuous east–north velocity command in $[-1, 1]^2$, followed by circular projection to respect the maximum speed.
Rollout length	n_{steps}	2048	Number of steps collected per environment before each PPO update.
Mini-batch size	B	256	Mini-batch size used during each gradient update.
Optimization epochs	N_{epochs}	10	Number of passes over each rollout.
Learning rate	α	3×10^{-4}	Adam learning rate used for the actor and critic.
Discount factor	γ	0.99	Reward discount factor.
GAE parameter	λ_{GAE}	0.95	Generalized advantage estimation applied to the team-average reward.
Policy clipping range	ϵ	0.2	Clipping range of the PPO probability ratio.
Value clipping range	ϵ_v	0.2	Clipping range of the value-function update.
Entropy coefficient	c_{ent}	0.01	Entropy regularization coefficient.
Value-loss coefficient	c_v	0.5	Weight of the value-function loss.
Maximum gradient norm	$g_{\max}$	0.5	Global gradient clipping threshold.
Training steps	T	2×10^6	Nominal environment steps per training run.
Independent seeds	–	0–9	Ten random seeds for each selected surface or underwater IPPO configuration.
Surface training forecast-error radius	R_{train}	0–800 km	The candidate surface formation-center radial offset is sampled uniformly in $[0, 800]$ km, with a uniformly sampled bearing. This models the assumed upper bound of TC track-forecast error; offshore-only rejection sampling is then applied as part of the safety layer.
Surface validation center offsets	R_{test}	100/200/400/600/800 km	Fixed surface formation-center offsets with random bearings, evaluated without retraining. These values represent increasing track-forecast displacement.
Underwater initialization radius	$R_{0,\text{u}}$	$\leq$200 km	At the end time of each surface segment, the separate subsurface episode initializes four vehicles with radial offsets sampled uniformly in $[0, 200]$ km and uniformly sampled bearings. Surface endpoint positions are not transferred. The 0–800 km forecast-error protocol is surface-only.
Decision interval	Δt	10 min	Environmental observation and action interval.
Surface formation radius	R_{ring}	100 km	Surface SUVs use equal angular spacing of $360°/N$ on a 100 km ring.
Surface fleet sizes	N	4/8/12/16	Number of SUVs in the final surface configurations.
Underwater fleet size	N	4	Number of SUVs in each underwater comparison case.
Underwater depths below the sea surface	z	0/10/20/30/40/50/60 m	Seven ocean-current observation depths used in the final subsurface matrix.
Underwater methods	–	IPPO; PF; A*; lawn; greedy	IPPO and four traditional planners evaluated on the same episodes and deployments.
Underwater duration	H	20 h	Fixed simulation horizon for the final subsurface comparison; this is not an endurance specification.
Surface ground-speed limits	$V^{\text{s}}_{\max}$	12 or 20 m/s	Two commanded ground-speed envelopes; these are modeled scenarios, not independently measured certifications.
Submerged speed limits	$V^{\text{u}}_{\max}$	1.5 or 2.0 m/s	Two propulsion-speed limits after unit-disk action projection used for underwater IPPO evaluation.
Compute device	–	CPU	All reported IPPO experiments were trained on CPU.

References

[1] T. Knutson, S. J. Camargo, J. C. L. Chan, et al., Tropical Cyclones and Climate Change Assessment: Part I: Detection and Attribution, Bulletin of the American Meteorological Society 100 (10) (2019) 1987–2007. doi:10.1175/BAMS-D-18-0189.1.

[2] T. Knutson, S. J. Camargo, J. C. L. Chan, et al., Tropical Cyclones and Climate Change Assessment: Part II: Projected Response to Anthropogenic Warming, Bulletin of the American Meteorological Society 101 (3) (2020) E303–E322. doi:10.1175/BAMS-D-18-0194.1.

[3] S. S. Chand, K. J. E. Walsh, S. J. Camargo, et al., Declining Tropical Cyclone Frequency under Global Warming, Nature Climate Change 12 (7) (2022) 655–661. doi:10.1038/s41558-022-01388-4.

[4] Y. Li, Y. Tang, S. Wang, et al., Recent Increases in Tropical Cyclone Rapid Intensification Events in Global Offshore Regions, Nature Communications 14 (1) (2023) 5167. doi:10.1038/s41467-023-40605-2.

[5] W. Mei, S.-P. Xie, Intensification of Landfalling Typhoons over the Northwest Pacific since the Late 1970s, Nature Geoscience 9 (10) (2016) 753–757. doi:10.1038/ngeo2792.

[6] L. Liu, Y. Wang, R. Zhan, et al., Increasing Destructive Potential of Landfalling Tropical Cyclones over China, Journal of Climate 33 (9) (2020) 3731–3743. doi:10.1175/JCLI-D-19-0451.1.

[7] K. Balaguru, C.-C. Chang, L. R. Leung, et al., A Global Increase in Nearshore Tropical Cyclone Intensification, Earth's Future 12 (5) (2024) e2023EF004230. doi:10.1029/2023EF004230.

[8] J. P. Kossin, K. A. Emanuel, S. J. Camargo, Past and Projected Changes in Western North Pacific Tropical Cyclone Exposure, Journal of Climate 29 (16) (2016) 5725–5739. doi:10.1175/JCLI-D-16-0076.1.

[9] J. P. Cangialosi, E. Blake, M. DeMaria, et al., Recent Progress in Tropical Cyclone Intensity Forecasting at the National Hurricane Center, Weather and Forecasting 35 (5) (2020) 1913–1922. doi:10.1175/WAF-D-20-0059.1.

[10] X. Huang, X. Peng, J. Fei, et al., Evaluation and Error Analysis of Official Tropical Cyclone Intensity Forecasts during 2005–2018 for the Western North Pacific, Journal of the Meteorological Society of Japan. Ser. II 99 (1) (2021) 139–163. doi:10.2151/jmsj.2021-008.

[11] H. Zhang, H. He, W.-Z. Zhang, et al., Upper Ocean Response to Tropical Cyclones: A Review, Geoscience Letters 8 (1) (2021) 1. doi:10.1186/s40562-020-00170-8.

[12] K. Balaguru, P. Chang, R. Saravanan, et al., Ocean Barrier Layers' Effect on Tropical Cyclone Intensification, Proceedings of the National Academy of Sciences 109 (36) (2012) 14343–14347. doi:10.1073/pnas.1201364109.

[13] M. A. Bourassa, T. Meissner, I. Cerovecki, et al., Remotely Sensed Winds and Wind Stresses for Marine Forecasting and Ocean Modeling, Frontiers in Marine Science 6 (2019) 443. doi:10.3389/fmars.2019.00443.

[14] B. W. Klotz, E. W. Uhlhorn, Improved Stepped Frequency Microwave Radiometer Tropical Cyclone Surface Winds in Heavy Precipitation, Journal of Atmospheric and Oceanic Technology 31 (11) (2014) 2392–2408. doi:10.1175/JTECH-D-14-00028.1.

[15] R. Domingues, A. Kuwano-Yoshida, P. Chardon-Maldonado, et al., Ocean Observations in Support of Studies and Forecasts of Tropical and Extratropical Cyclones, Frontiers in Marine Science 6 (2019) 446. doi:10.3389/fmars.2019.00446.

[16] J. D. Doyle, J. R. Moskaitis, J. W. Feldmeier, et al., A View of Tropical Cyclones from Above: The Tropical Cyclone Intensity Experiment, Bulletin of the American Meteorological Society 98 (10) (2017) 2113–2134. doi:10.1175/BAMS-D-16-0055.1.

[17] J. J. Cione, et al., Eye of the Storm: Observing Hurricanes with a Small Unmanned Aircraft System, Bulletin of the American Meteorological Society 101 (2) (2020) E186–E205. doi:10.1175/BAMS-D-19-0169.1.

[18] D. L. Rudnick, Ocean Research Enabled by Underwater Gliders, Annual Review of Marine Science 8 (1) (2016) 519–541. doi:10.1146/annurev-marine-122414-033913.

[19] J. Hwang, N. Bose, S. Fan, AUV Adaptive Sampling Methods: A Review, Applied Sciences 9 (15) (2019) 3145. doi:10.3390/app9153145.

[20] J. Dong, R. Domingues, G. Goni, et al., Impact of Assimilating Underwater Glider Data on Hurricane Gonzalo (2014) Forecasts, Weather and Forecasting 32 (3) (2017) 1143–1159. doi:10.1175/WAF-D-16-0182.1.

[21] G. R. Halliwell, G. J. Goni, M. F. Mehari, et al., OSSE Assessment of Underwater Glider Arrays to Improve Ocean Model Initialization for Tropical Cyclone Prediction, Journal of Atmospheric and Oceanic Technology 37 (3) (2020) 467–487. doi:10.1175/JTECH-D-18-0195.1.

[22] X. Chen, F. Zhang, Development of a Convection-Permitting Air–Sea-Coupled Ensemble Data Assimilation System for Tropical Cyclone Prediction, Journal of Advances in Modeling Earth Systems 11 (11) (2019) 3474–3496. doi:10.1029/2019MS001795.

[23] L. Phillipson, Y. Li, R. Toumi, Strongly Coupled Assimilation of a Hypothetical Ocean Current Observing Network within a Regional Ocean–Atmosphere Coupled Model: An OSSE Case Study of Typhoon Hato, Monthly Weather Review 149 (5) (2021) 1317–1336. doi:10.1175/MWR-D-20-0108.1.

[24] L. Liu, A. Mehra, D. Kleist, et al., Impact of Assimilating Satellite and Glider Observations on Hurricane Isaias (2020) Forecast Using Marine JEDI, Weather and Forecasting 38 (9) (2023) 1807–1826. doi:10.1175/WAF-D-22-0014.1.

[25] S. Chen, J. A. Cummings, J. M. Schmidt, et al., Targeted Ocean Sampling Guidance for Tropical Cyclones, Journal of Geophysical Research: Oceans 122 (5) (2017) 3505–3518. doi:10.1002/2017JC012727.

[26] M. Mu, F. Zhou, H. Wang, A Method for Identifying the Sensitive Areas in Targeted Observations for Tropical Cyclone Prediction: Conditional Nonlinear Optimal Perturbation, Monthly Weather Review 137 (5) (2009) 1623–1639. doi:10.1175/2008MWR2640.1.

[27] C.-C. Wu, J.-H. Chen, P.-H. Lin, et al., Targeted Observations of Tropical Cyclone Movement Based on the Adjoint-Derived Sensitivity Steering Vector, Journal of the Atmospheric Sciences 64 (7) (2007) 2611–2626. doi:10.1175/JAS3974.1.

[28] C. Dong, G. Wu, Y. Zhang, H. Zhang, Q. Wang, An Innovative Submersible Unmanned Vessel: Blue Whale, Ocean-Land-Atmosphere Research 4 (2025) 0102. doi:10.34133/olar.0102.

[29] Y. Tian, W. Li, Z. Wang, Development and Validation of a Wave-Propelled Semi-Submersible Unmanned Vehicle, Polish Maritime Research 31 (4) (2024) 31–42. doi:10.2478/pomr-2024-0048.

[30] H. Zhou, K. Liu, H. Xu, X. Feng, Experimentally Verified Depth Control of an Unmanned Semi-Submersible Vehicle, IEEE Access 7 (2019) 94254–94262. doi:10.1109/ACCESS.2019.2925149.

[31] H. Chen, J. Li, Y. Xuan, et al., First Rocketsonde Launched from an Unmanned Semi-Submersible Vehicle, Advances in Atmospheric Sciences 36 (4) (2019) 339–345. doi:10.1007/s00376-018-8249-5.

[32] Shanghai Jiao Tong University, Video: Toward the Far Seas and Deep Ocean: The "Far-Seas Floating Island" Supports Stable Marine Research, Shanghai Jiao Tong University News, published 28 March 2026 (2026).
URL https://news.sjtu.edu.cn/mtjj/20260329/220899.html

[33] C. M. Wang, Z. Y. Tay, Very Large Floating Structures: Applications, Research and Development, Procedia Engineering 14 (2011) 62–72. doi:10.1016/j.proeng.2011.07.007.

[34] P. Palo, Mobile Offshore Base: Hydrodynamic Advancements and Remaining Challenges, Marine Structures 18 (2) (2005) 133–147. doi:10.1016/j.marstruc.2005.07.007.

[35] P. E. Hart, N. J. Nilsson, B. Raphael, A Formal Basis for the Heuristic Determination of Minimum Cost Paths, IEEE Transactions on Systems Science and Cybernetics 4 (2) (1968) 100–107. doi:10.1109/TSSC.1968.300136.

[36] S. Koenig, M. Likhachev, D* Lite, in: Eighteenth National Conference on Artificial Intelligence, 2002, pp. 476–483.

[37] S. Karaman, E. Frazzoli, Sampling-Based Algorithms for Optimal Motion Planning, The International Journal of Robotics Research 30 (7) (2011) 846–894. doi:10.1177/0278364911406761.

[38] O. Khatib, Real-Time Obstacle Avoidance for Manipulators and Mobile Robots, The International Journal of Robotics Research 5 (1) (1986) 90–98. doi:10.1177/027836498600500106.

[39] D. Q. Mayne, J. B. Rawlings, C. V. Rao, et al., Constrained Model Predictive Control: Stability and Optimality, Automatica 36 (6) (2000) 789–814. doi:10.1016/S0005-1098(99)00214-9.

[40] M. Panda, B. Das, B. Subudhi, et al., A Comprehensive Review of Path Planning Algorithms for Autonomous Underwater Vehicles, International Journal of Automation and Computing 17 (3) (2020) 321–352. doi:10.1007/s11633-019-1204-9.

[41] P. F. Christiano, J. Leike, T. Brown, et al., Deep Reinforcement Learning from Human Preferences, in: Advances in Neural Information Processing Systems, Vol. 30, 2017, pp. 4299–4307.

[42] L. Ouyang, J. Wu, X. Jiang, et al., Training Language Models to Follow Instructions with Human Feedback, in: Advances in Neural Information Processing Systems, Vol. 35, 2022, pp. 27730–27744. doi:10.52202/068431-2011.

[43] D. Guo, D. Yang, H. Zhang, et al., DeepSeek-R1 Incentivizes Reasoning in LLMs through Reinforcement Learning, Nature 645 (8081) (2025) 633–638. doi:10.1038/s41586-025-09422-z.

[44] J. Hwangbo, J. Lee, A. Dosovitskiy, et al., Learning Agile and Dynamic Motor Skills for Legged Robots, Science Robotics 4 (26) (2019) eaau5872. doi:10.1126/scirobotics.aau5872.

[45] M. Andrychowicz, B. Baker, M. Chociej, et al., Learning Dexterous In-Hand Manipulation, The International Journal of Robotics Research 39 (1) (2020) 3–20. doi:10.1177/0278364919887447.

[46] E. Kaufmann, L. Bauersfeld, A. Loquercio, et al., Champion-Level Drone Racing Using Deep Reinforcement Learning, Nature 620 (7976) (2023) 982–987. doi:10.1038/s41586-023-06419-4.

[47] O. Vinyals, I. Babuschkin, W. M. Czarnecki, et al., Grandmaster Level in StarCraft II Using Multi-Agent Reinforcement Learning, Nature 575 (7782) (2019) 350–354. doi:10.1038/s41586-019-1724-z.

[48] C. Berner, G. Brockman, B. Chan, et al., Dota 2 with Large Scale Deep Reinforcement Learning (2019). arXiv:1912.06680, doi:10.48550/arXiv.1912.06680.

[49] C. S. de Witt, T. Gupta, D. Makoviichuk, et al., Is Independent Learning All You Need in the StarCraft Multi-Agent Challenge? (2020). arXiv:2011.09533, doi:10.48550/arXiv.2011.09533.

[50] C. Yu, A. Velu, E. Vinitsky, et al., The Surprising Effectiveness of PPO in Cooperative Multi-Agent Games, in: Advances in Neural Information Processing Systems, Vol. 35, 2022, pp. 24611–24624. doi:10.52202/068431-1787.

[51] M. Samvelyan, T. Rashid, C. S. de Witt, et al., The StarCraft Multi-Agent Challenge, in: Proceedings of the 18th International Conference on Autonomous Agents and MultiAgent Systems, International Foundation for Autonomous Agents and Multiagent Systems, 2019, pp. 2186–2188. doi:10.65109/lvzz5205.

[52] E. Fiorelli, N. E. Leonard, P. Bhatta, et al., Multi-AUV Control and Adaptive Sampling in Monterey Bay, IEEE Journal of Oceanic Engineering 31 (4) (2006) 935–948. doi:10.1109/JOE.2006.880429.

[53] I. Masmitja, M. Martin, T. O'Reilly, et al., Dynamic Robotic Tracking of Underwater Targets Using Reinforcement Learning, Science Robotics 8 (80) (2023) eade7811. doi:10.1126/scirobotics.ade7811.

[54] S. Wang, C. Lin, G. Han, et al., Multi-AUV Cooperative Underwater Multi-Target Tracking Based on Dynamic-Switching-Enabled Multi-Agent Reinforcement Learning, IEEE Transactions on Mobile Computing 24 (5) (2025) 4296–4311. doi:10.1109/TMC.2024.3521889.

[55] Z. Zhao, Q. Hu, H. Feng, et al., A Cooperative Hunting Method for Multi-AUV Swarm in Underwater Weak Information Environment with Obstacles, Journal of Marine Science and Engineering 10 (9) (2022) 1266. doi:10.3390/jmse10091266.

[56] X. Li, P. Dong, H. Tao, et al., A Multi-AUV Cooperative Search Scheme Based on Acoustic-Optical Communication and Deep Reinforcement Learning, in: 2024 IEEE International Conference on High Performance Computing and Communications, 2024, pp. 785–790. doi:10.1109/HPCC64274.2024.00108.

[57] F. Qi, X. Ma, J. Yang, Tropical Cyclone Tracking for Autonomous Underwater Vehicles Based on Forecast Path Correction Model, Ocean Engineering 293 (2024) 116768. doi:10.1016/j.oceaneng.2024.116768.

[58] F. Qi, X. Ma, Y. Ding, J. Yang, Unmanned Active Observation Planning of Near-Surface Ocean under Tropical Cyclone Condition, Journal of Ocean Engineering and Science 11 (6) (2026) 2077–2088. doi:10.1016/j.joes.2026.05.012.

[59] J. Schulman, F. Wolski, P. Dhariwal, et al., Proximal Policy Optimization Algorithms (2017). arXiv:1707.06347, doi:10.48550/arXiv.1707.06347.

[60] H. Hersbach, B. Bell, P. Berrisford, et al., The ERA5 Global Reanalysis, Quarterly Journal of the Royal Meteorological Society 146 (730) (2020) 1999–2049. doi:10.1002/qj.3803.

[61] E. P. Chassignet, H. E. Hurlburt, O. M. Smedstad, et al., The HYCOM (HYbrid Coordinate Ocean Model) Data Assimilative System, Journal of Marine Systems 65 (1–4) (2007) 60–83. doi:10.1016/j.jmarsys.2005.09.016.

[62] K. R. Knapp, M. C. Kruk, D. H. Levinson, et al., The International Best Track Archive for Climate Stewardship (IBTrACS): Unifying Tropical Cyclone Data, Bulletin of the American Meteorological Society 91 (3) (2010) 363–376. doi:10.1175/2009BAMS2755.1.